

Gravitational wave multimessenger astronomy in the 2030s

Antonella Palmese

4 August 2026

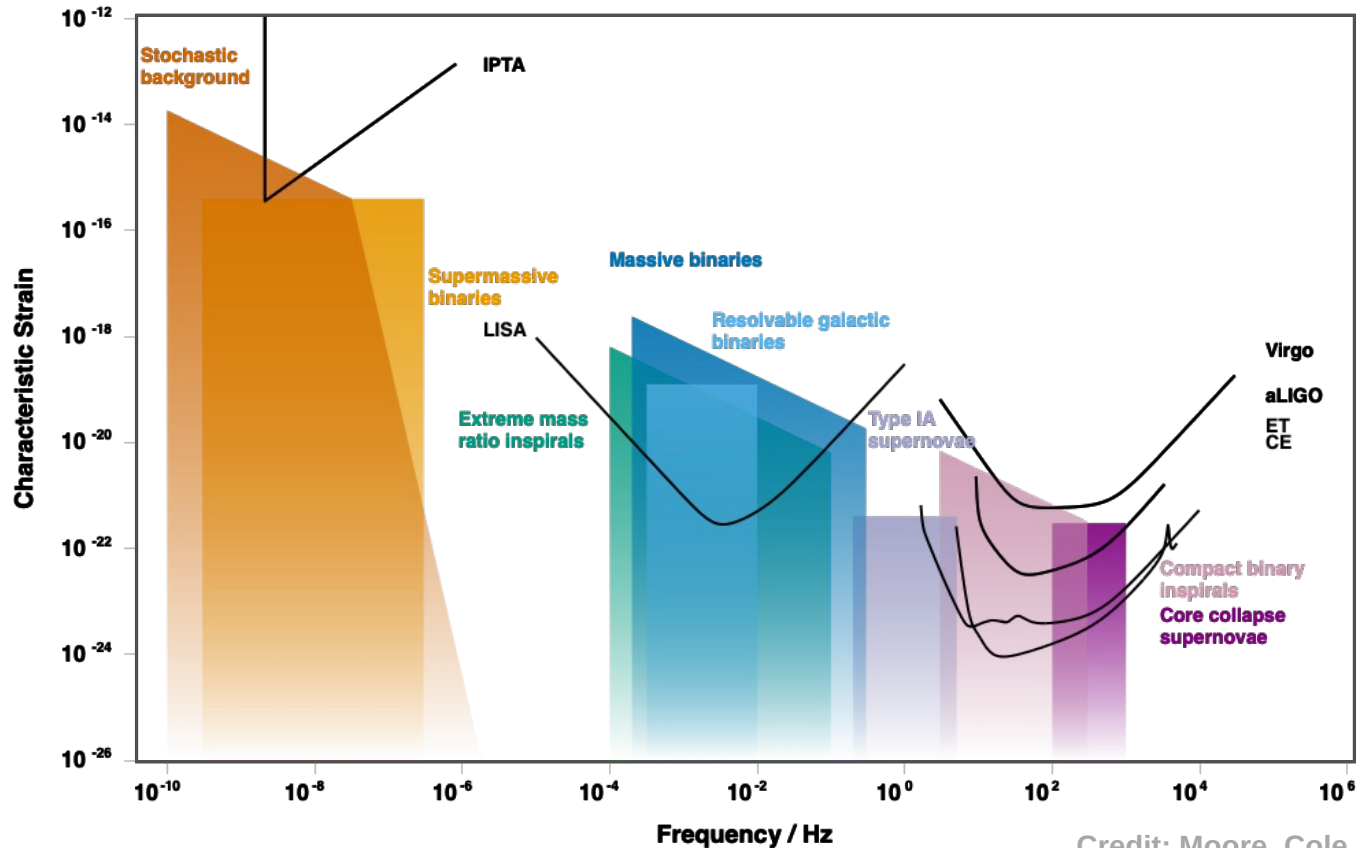
Rubin 2036 workshop

SLAC



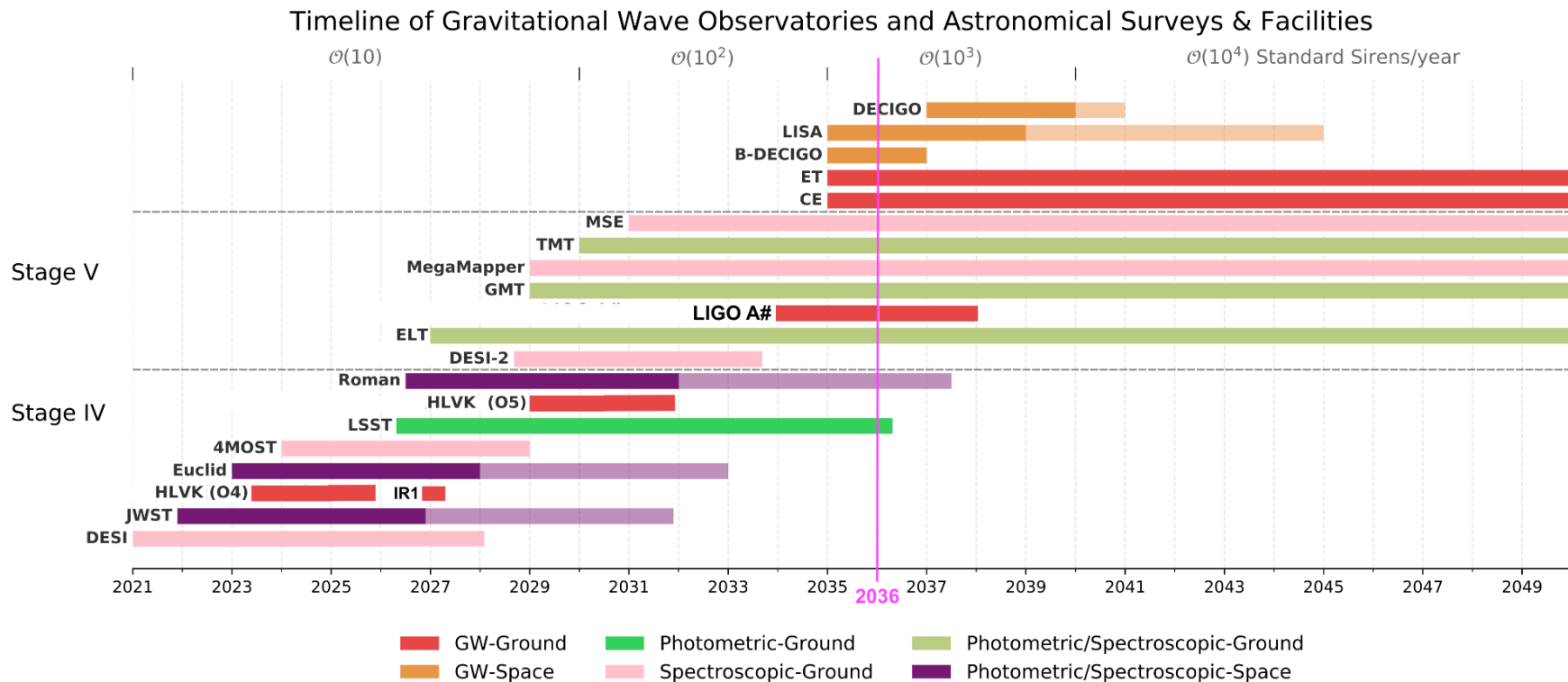
McWilliams Center for
Cosmology & Astrophysics

Gravitational wave detectors



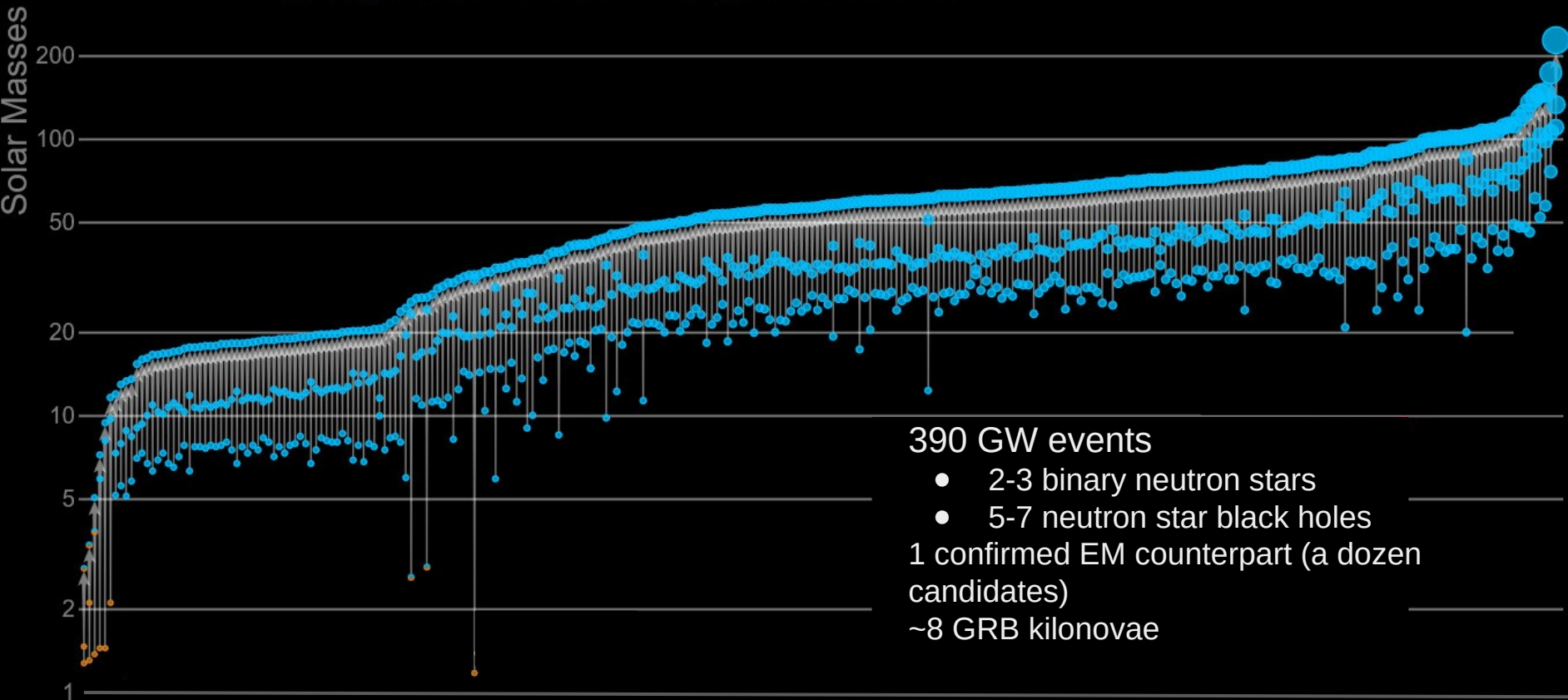
Credit: Moore, Cole & Berry

Timeline

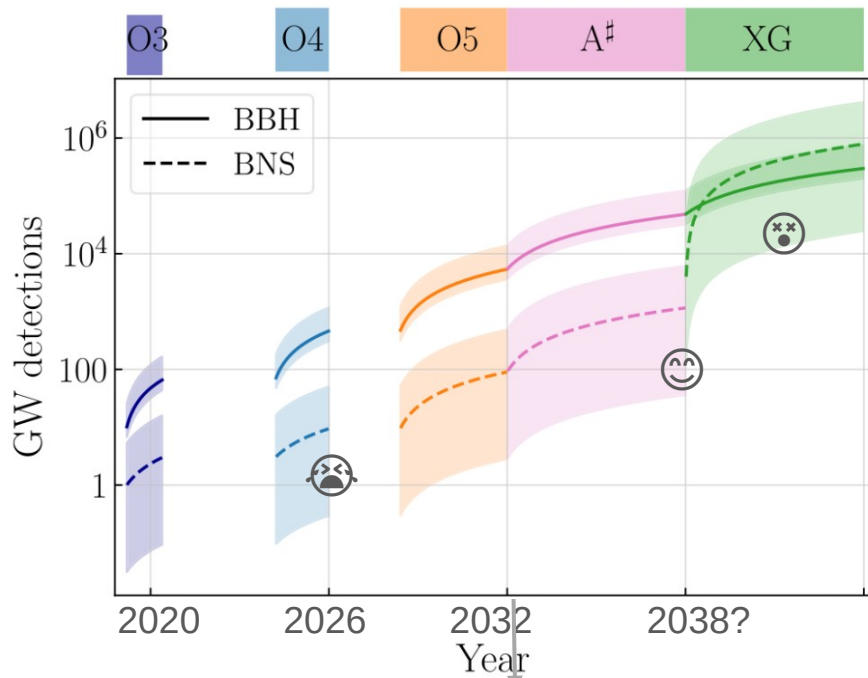


Masses in the Stellar Graveyard

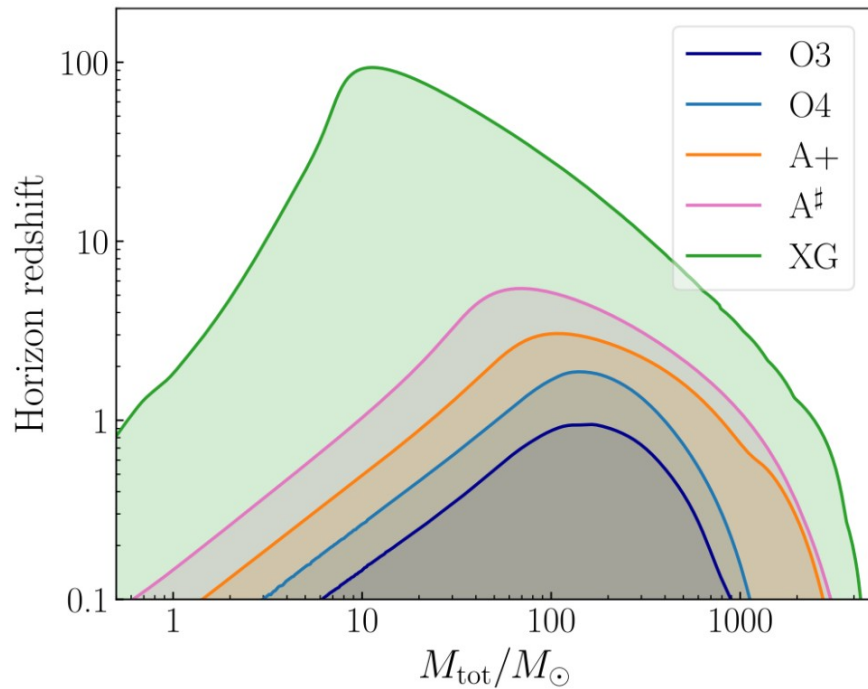
LIGO-Virgo-KAGRA Black Holes LIGO-Virgo-KAGRA Neutron Stars



Detection rates and horizons

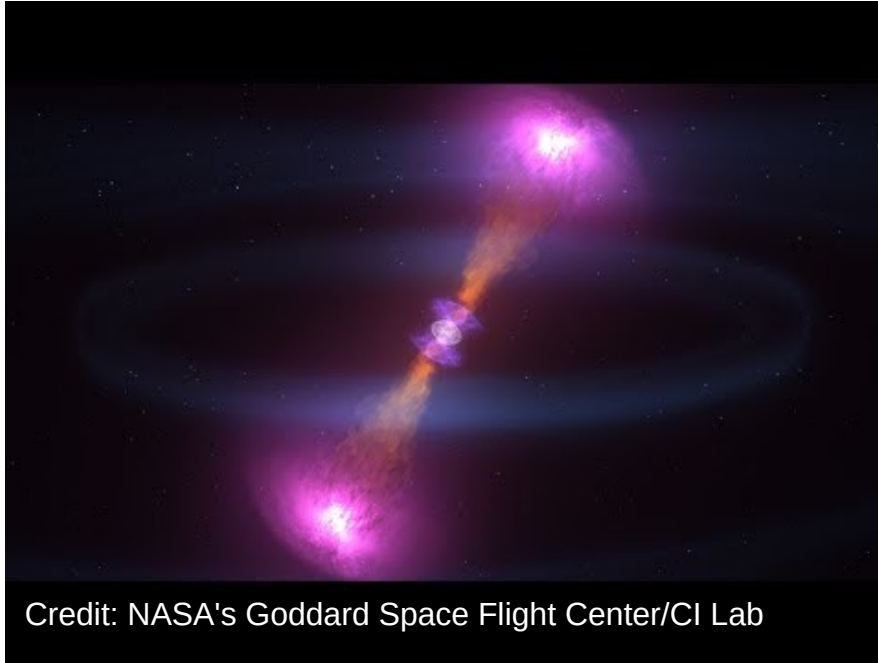


Maybe 1 NSBH counterpart
Kunnumkai et al. 2025



Chen, Ezquiaga & Gupta 2024 (Adapted)

Binary Neutron star mergers EM counterparts

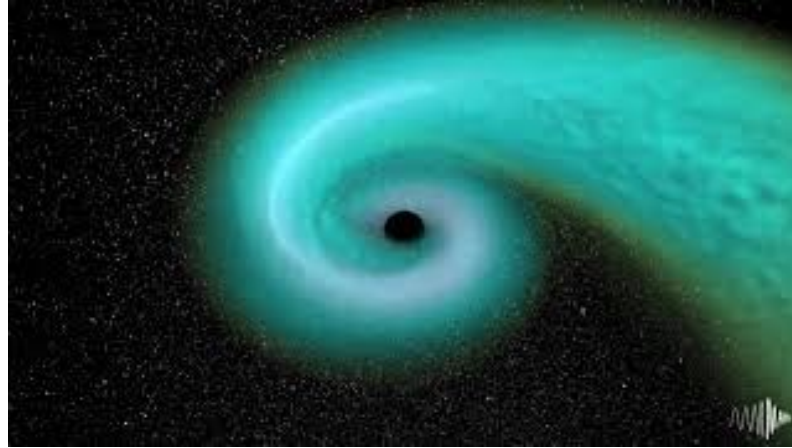


Credit: NASA's Goddard Space Flight Center/CI Lab

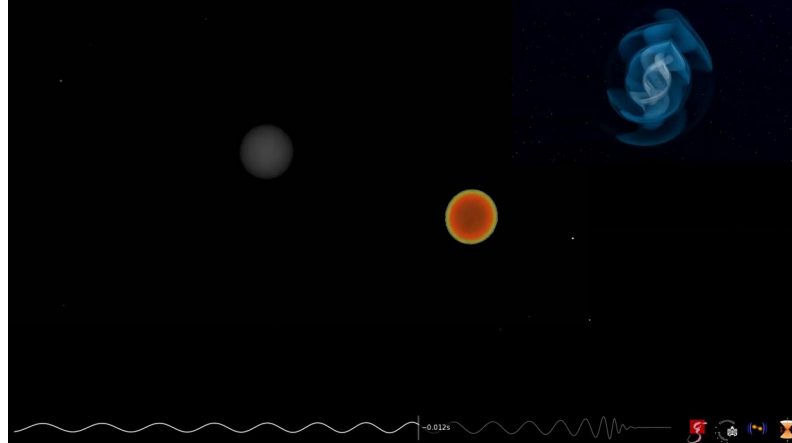
- **Relativistic jet** - gamma rays (magenta)
- Ejecta from **kilonova** - UV/optical/infrared (violet and white to red)
- **Afterglow** - X-rays, optical, radio

Neutron Star-black hole mergers EM counterparts

EM Counterpart

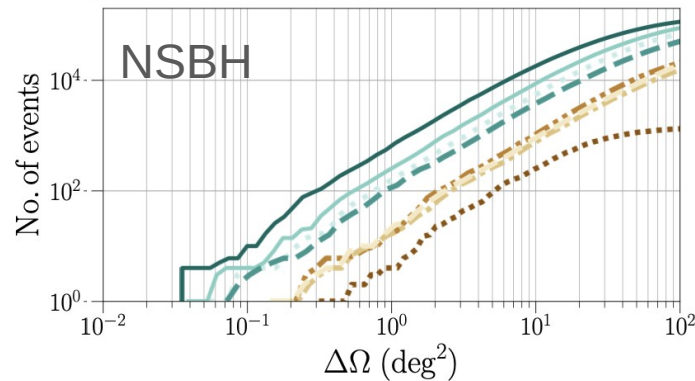
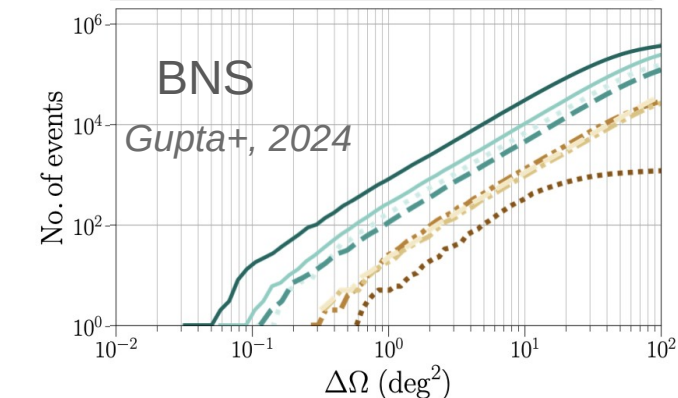
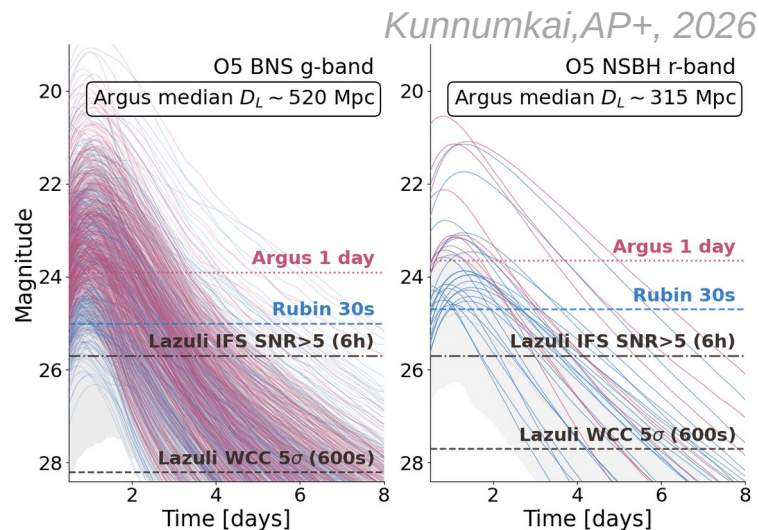


No EM Counterpart

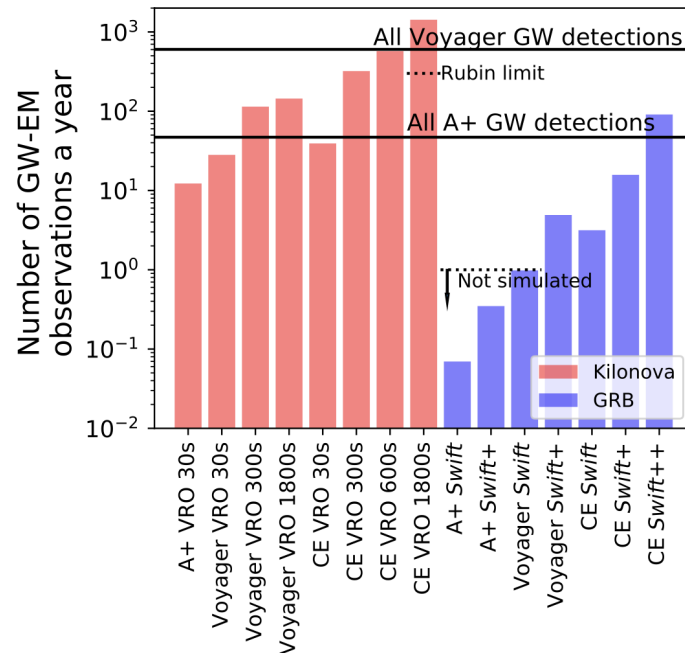
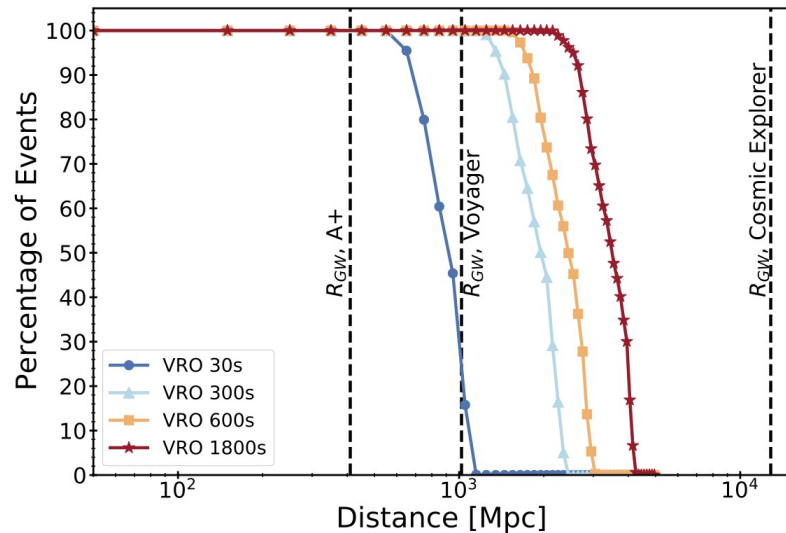


Multimessenger kilonova detections prospects

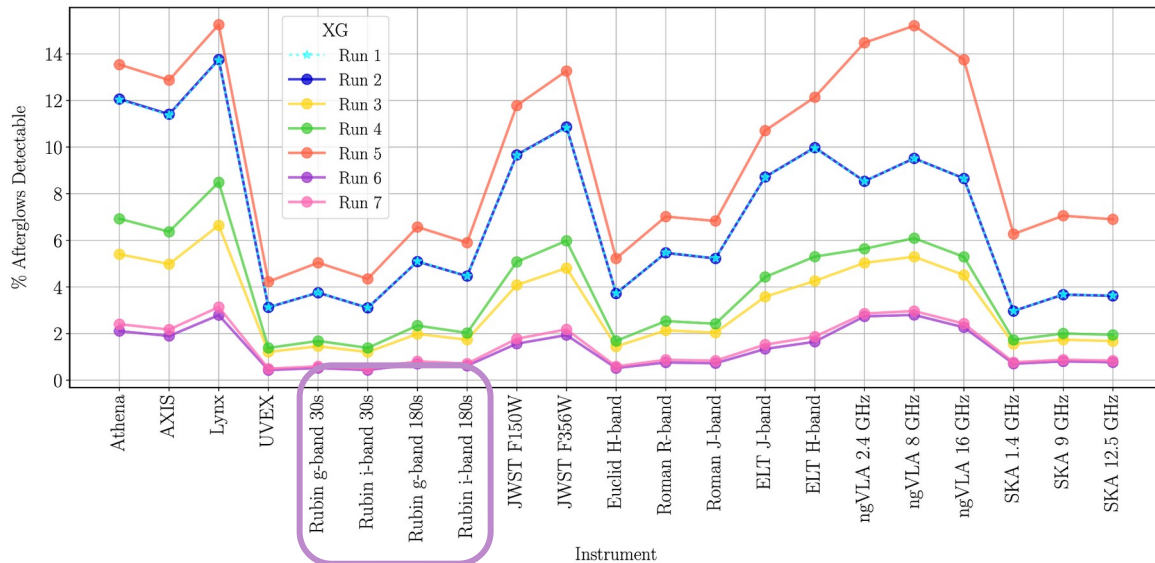
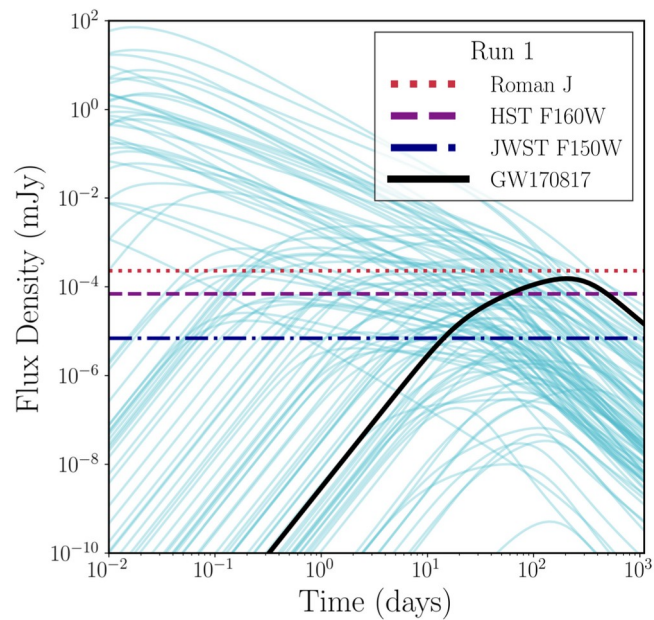
Rubin's wide FoV and sensitivity are crucial



Multimessenger kilonova detections prospects



Multimessenger afterglow detections prospects

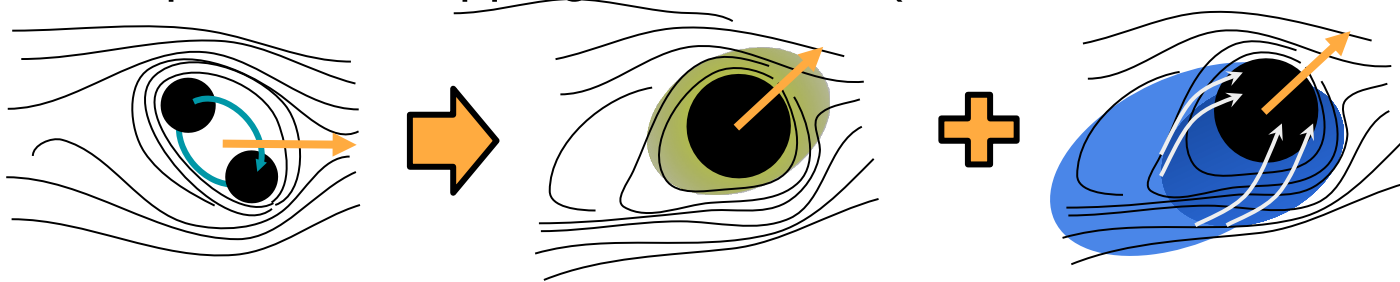


Still tens of detections per year!

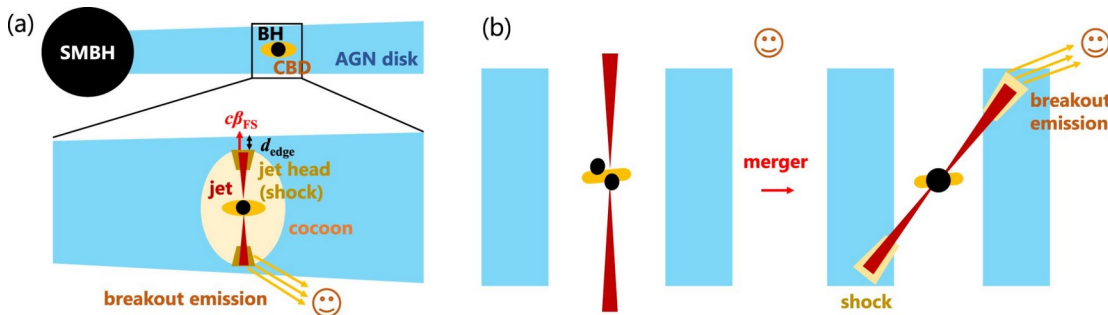
EM emission from BBH in AGN disks

See also Bartos+17

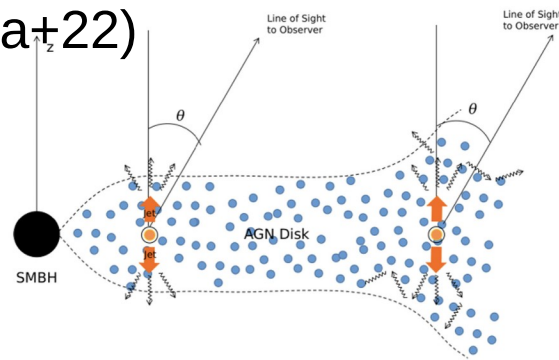
Ram-pressure stripping + Bondi tail (McKernan+19, Antoni+19)



Jet breakout, shock cooling (Tagawa+23, 24)



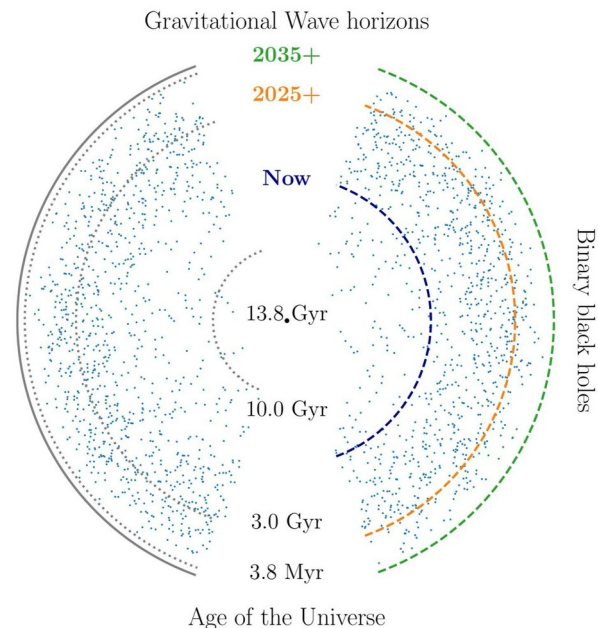
Jet afterglow (~Wang, Lazzati & Perna+22)



Science Cases - Standard Sirens

1. Compact binaries merge at **cosmological distances**, and we will be able to observe them all the way to the first mergers
2. GWs **travel unaltered** through the Universe, except for gravitational lensing
3. GW wavelengths are of **astrophysical scale**
4. Signals are understood **from first principles** → distance measurement is too!

Sub-percent H_0 precision with XG+Rubin
Other cosmological parameters constraints not competitive on their own, but interesting when combined with other probes
(*bright sirens*)



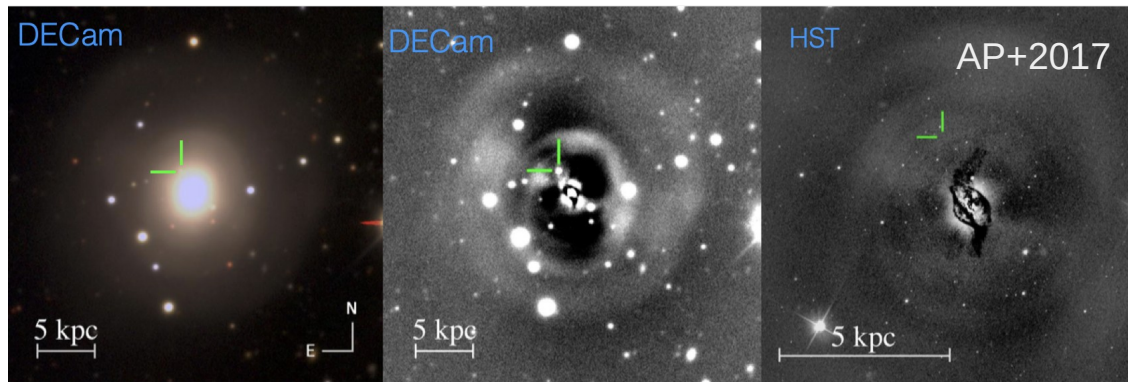
Chen, Ezquiaga & Gupta 2024

Science Cases - heavy elements production in kilonovae

H1 Hydrogen	 The big bang  Dying low-mass stars  White dwarf supernovae  Radioactive decay																He2 Helium															
Li3 Lithium	Be4 Beryllium	 Cosmic ray collisions  Dying high-mass stars  Merging neutron stars  Human-made																B5 Boron	C6 Carbon	N7 Nitrogen	O8 Oxygen	F9 Fluorine	Ne10 Neon									
Na11 Sodium	Mg12 Magnesium																	Al13 Aluminum	Si14 Silicon	P15 Phosphorus	S16 Sulfur	Cl17 Chlorine	Ar18 Argon									
K19 Potassium	Ca20 Calcium	Sc21 Scandium	Ti22 Titanium	V23 Vanadium	Cr24 Chromium	Mn25 Manganese	Fe26 Iron	Co27 Cobalt	Ni28 Nickel	Cu29 Copper	Zn30 Zinc	Ga31 Gallium	Ge32 Germanium	As33 Arsenic	Se34 Selenium	Br35 Bromine	Kr36 Krypton															
Rb37 Rubidium	Sr38 Strontium	Y39 Yttrium	Zr40 Zirconium	Nb41 Niobium	Mo42 Molybdenum	Tc43 Technetium	Ru44 Ruthenium	Rh45 Rhodium	Pd46 Palladium	Ag47 Silver	Cd48 Cadmium	In49 Indium	Sn50 Tin	Sb51 Antimony	Te52 Tellurium	I53 Iodine	Xe54 Xenon															
Cs55 Cesium	Ba56 Barium																	Hf72 Hafnium	Ta73 Tantalum	W74 Tungsten	Re75 Rhenium	Os76 Osmium	Ir77 Iridium	Pt78 Platinum	Au79 Gold	Hg80 Mercury	Tl81 Thallium	Pb82 Lead	Bi83 Bismuth	Po84 Polonium	At85 Astatine	Rn86 Radon
Fr87 Francium	Ra88 Radium																	Rf104 Rutherfordium	Db105 Dubnium	Sg106 Seaborgium	Bh107 Bohrium	Hs108 Hassium	Mt109 Meitnerium	Ds110 Darmstadtium	Rg111 Roentgenium	Cn112 Copernicium	Nh113 Nihonium	Fl114 Flerovium	Mc115 Moscovium	Lv116 Livermorium	Ts117 Tennessine	Og118 Oganesson
		La57 Lanthanum	Ce58 Cerium	Pr59 Praseodymium	Nd60 Neodymium	Pm61 Promethium	Sm62 Samarium	Eu63 Europium	Gd64 Gadolinium	Tb65 Terbium	Dy66 Dysprosium	Ho67 Holmium	Er68 Erbium	Tm69 Thulium	Yb70 Ytterbium	Lu71 Lutetium																
		Ac89 Actinium	Th90 Thorium	Pa91 Protactinium	U92 Uranium	Np93 Neptunium	Pu94 Plutonium	Am95 Americium	Cm96 Curium	Bk97 Berkelium	Cf98 Californium	Es99 Einsteinium	Fm100 Fermium	Md101 Mendelevium	No102 Nobelium	Lr103 Lawrencium																

Science Cases - Environments and formation channels

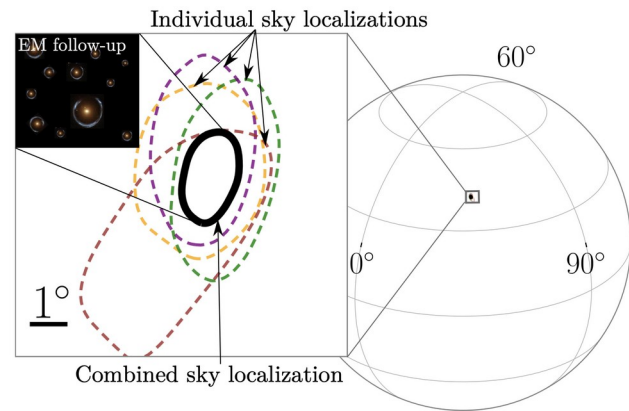
- **O(10) multimessenger events** to discern between formation channels that trace stellar mass, halo mass, SFR (Adhikari+20)
- **Morphology studies:** Galaxy merger-driven formation pathways may play a significant role (AP+2017; Skobe+2026)
- **Offsets and delay times** inform formation channels



Other gravitational wave cosmology-related cases

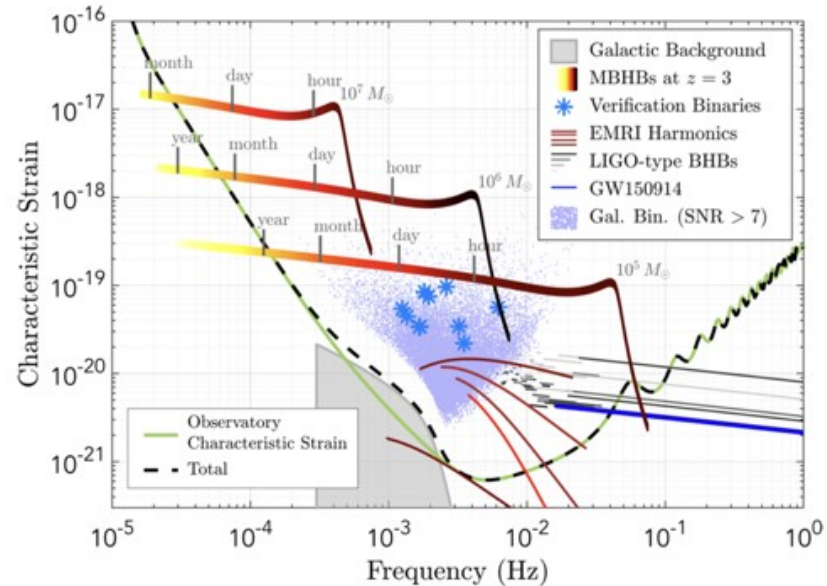
- **Cosmological Parameters**
 - Peculiar velocities
 - LSS Cross-correlations for bias and cosmological parameters
- **Modified Gravity**
 - Luminosity distance vs GW distance
- **Dark Matter**
 - Gravitational lensing of gravitational waves
 - Primordial black holes, exotic compact objects

Also consider: NS equation of state, unmodeled sources and new discoveries, black hole seeding and evolution with LISA



Hannuksela+2020

- Massive Black Hole Binaries
 - EM bright or dark, out to $z < 10$
- Stellar black holes
 - Expected dark
- Extreme Mass Ratio Inspirals
 - bright? May relate to TDEs/QPEs
- Galactic binaries



LISA massive black hole binaries - Rubin prospects

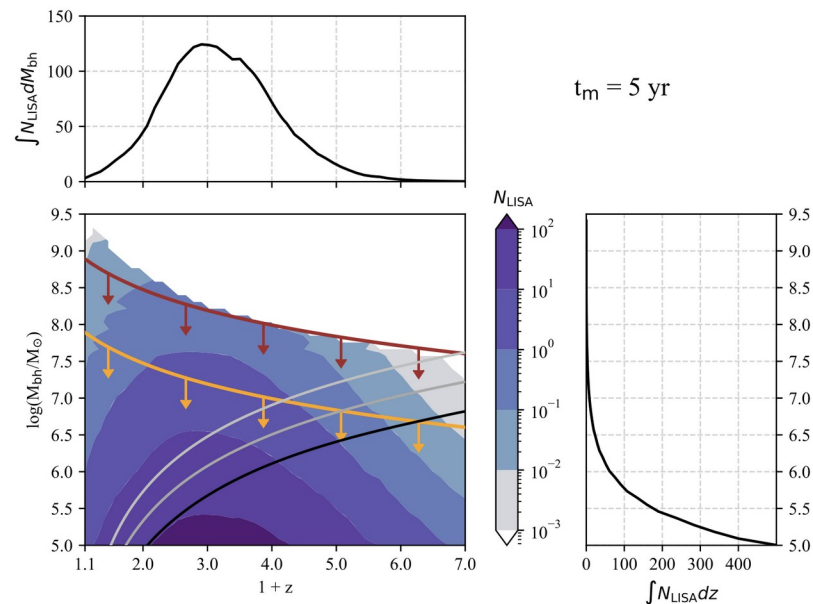
10-150 “verification” binaries detectable in Rubin and later in LISA (*Xin & Haiman 2021*)

Identifying the EM counterpart may be possible at 1 week/month before merger with Rubin candidates (*Xin & Haiman 2024*), enabling multiwavelength/pointed follow up

LISA sky localizations: 0.1–1000 deg²

Mass/z not enough for identification

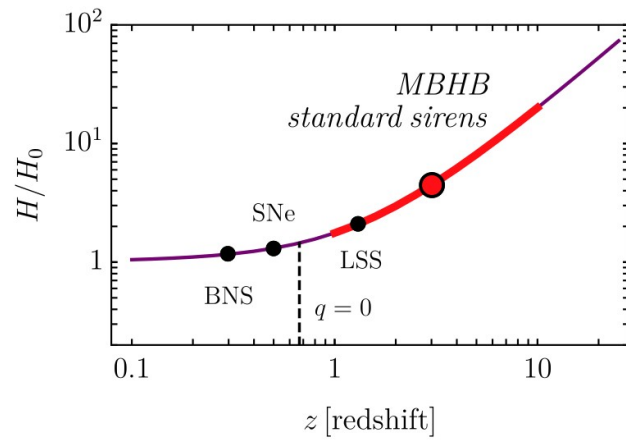
Continued monitoring through the LISA lifetime is ideal also to catch “Changing-look inspirals” (*Zrake et al. 2025*)



Xin & Haiman 2021, MNRAS

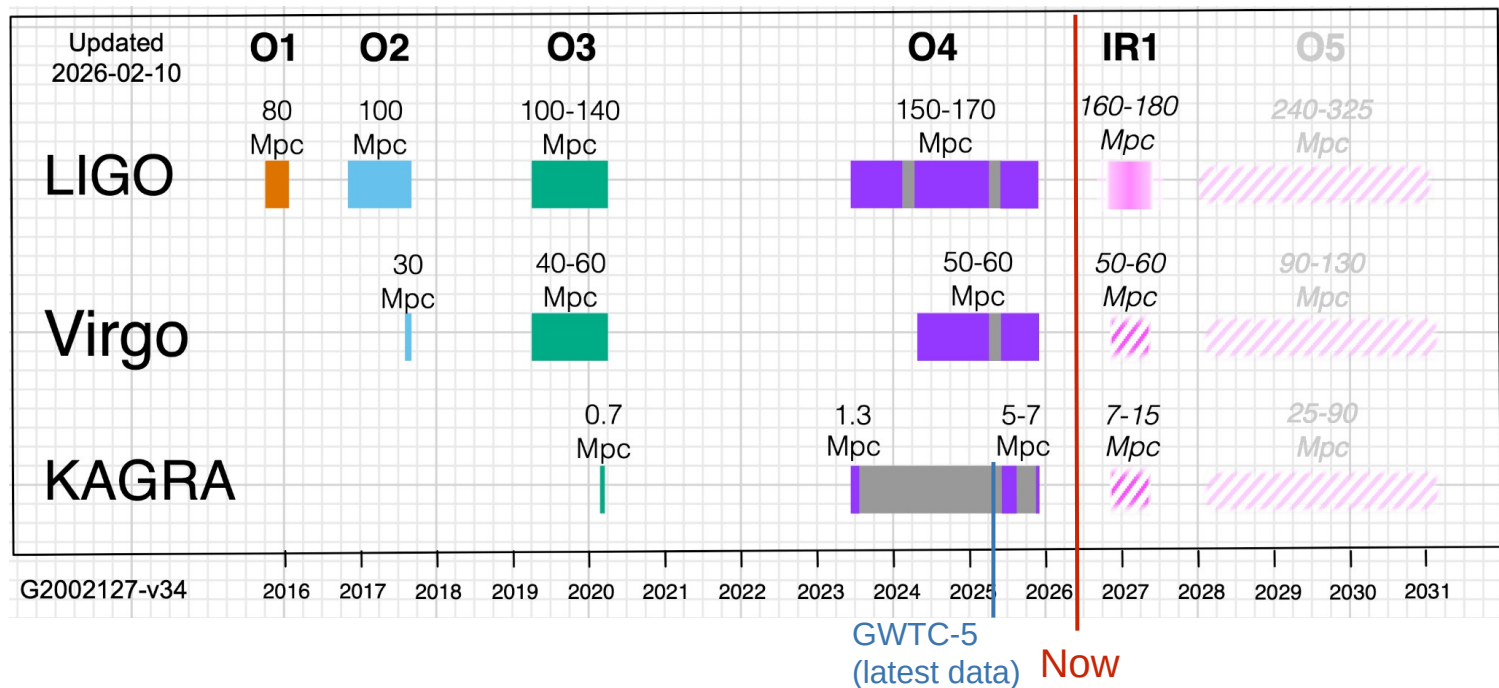
Conclusions

- **By 2036:**
 - We may not have a large (100s) sample of multimessenger events yet
 - BNS Rates are uncertain but we know GRB rates
 - Rubin may have identified candidate MBHBs, possibly not confirmed by LISA yet
- **Opportunities for Rubin from 2036:**
 - Contribute to precision cosmology and population studies with a large dedicated multimessenger program
 - A lot of other science cases not discussed here
 - Some of the most interesting results from GW astronomy came/may come from what we did not expect - a facility with a dynamic strategy is crucial
 - Coordination between GW detectors and Rubin+EM facilities (e.g. timelines). PI-led campaigns will become less and less sustainable in the XG era.



Caldwell+
2019

Current generation GW detectors timeline



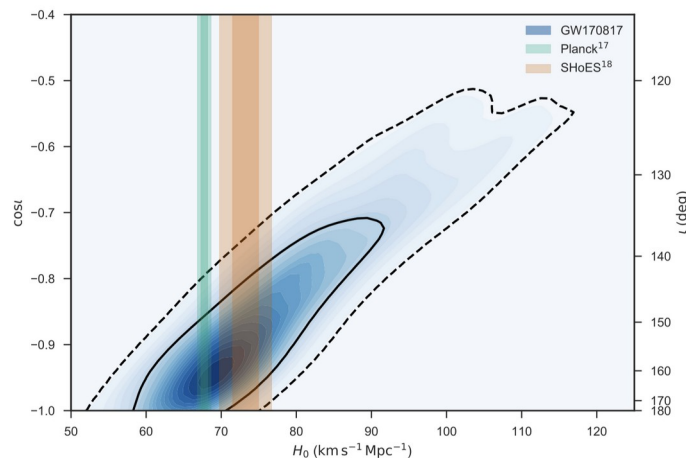
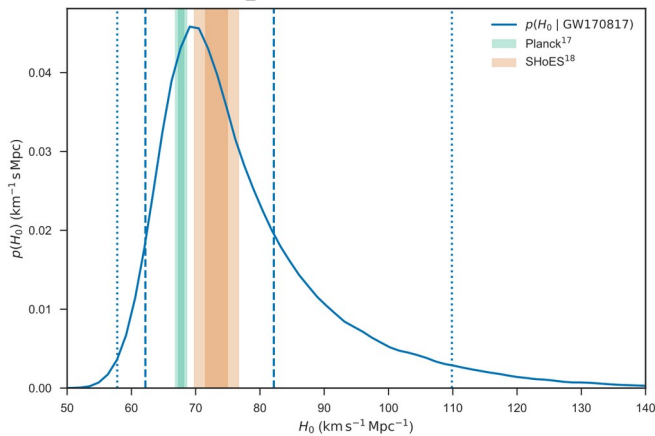
A#?



Binary neutron star mergers

GW170817 bright standard siren

LVC, EM partners+ 2017, Nature



- NGC 4993 ($z \sim 0.01$)

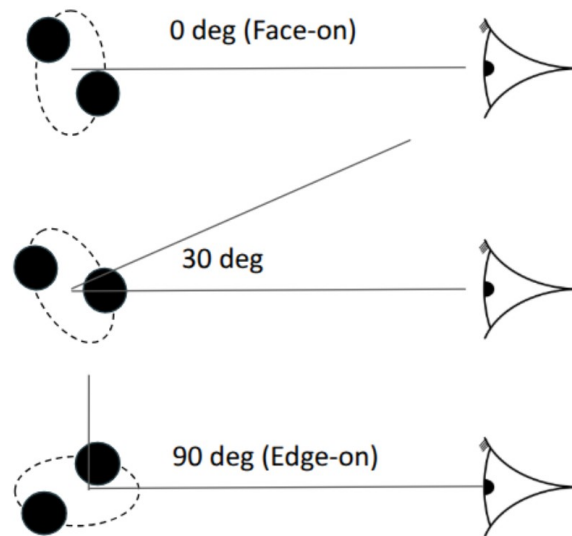
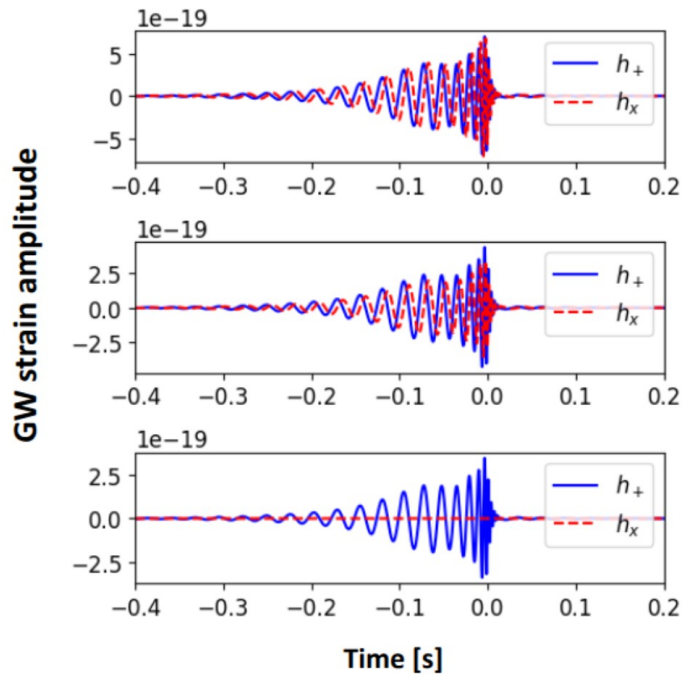
$$H_0 = 70_{-8}^{+12} \text{ km s}^{-1} \text{ Mpc}^{-1}$$

- **Ideal to understand Hubble constant tension:**

1. Self-calibrating: Independent of distance ladder
2. Cosmological model independent

- (Some) limitations:
 - Peculiar velocity
 - Inclination angle is correlated with luminosity distance

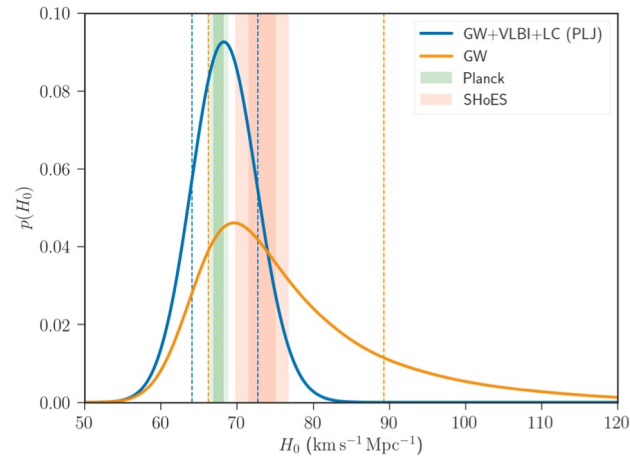
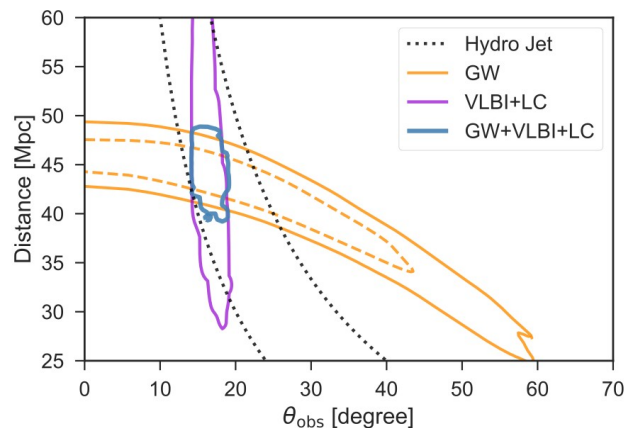
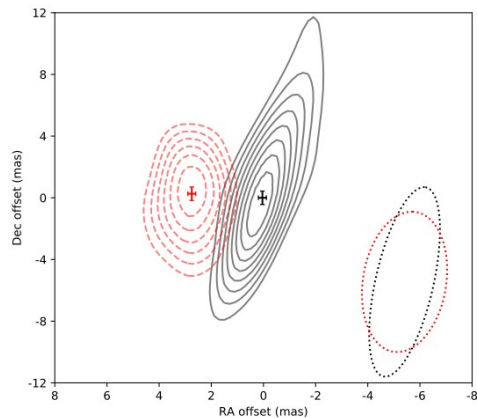
Standard sirens - viewing angle-distance degeneracy



Credit: S. Mastrogiovanni

GW170817 bright standard siren - viewing angle

Break **distance-inclination degeneracy** by constraining viewing angle from EM observations

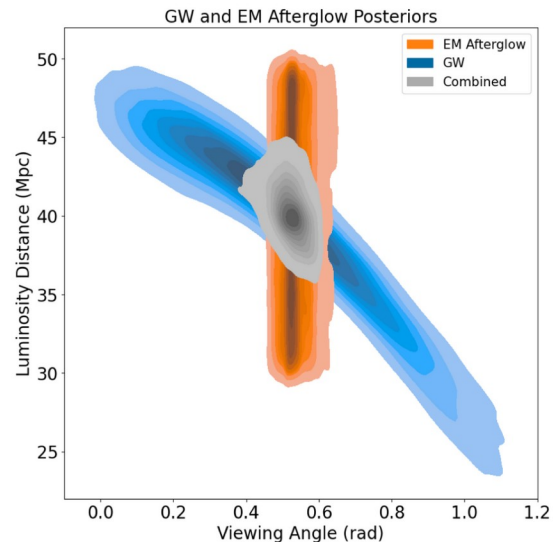
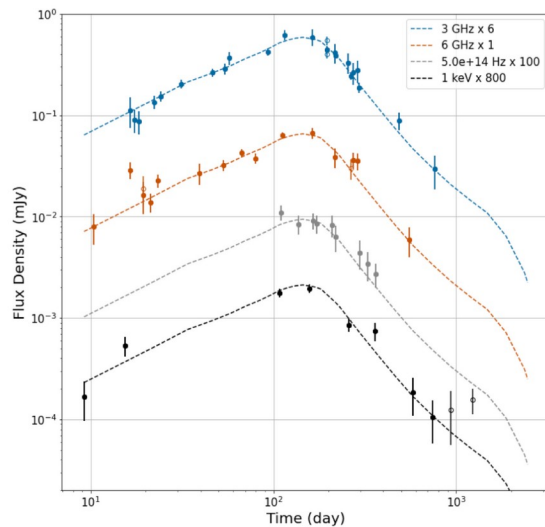


Mooley et al. 2018, Nature

Hotokezaka+2019, Nat. Ast.

GW170817 bright standard siren - viewing angle

- **GW170817 Afterglow at ~3.5 years:** including data beyond peak brightness
- Hydrodynamic simulation-based JetFit (Wu & MacFadyen 2018) to model the afterglow, consistently **from X-ray to radio**, with simulations from McDowell & MacFadyen 2023

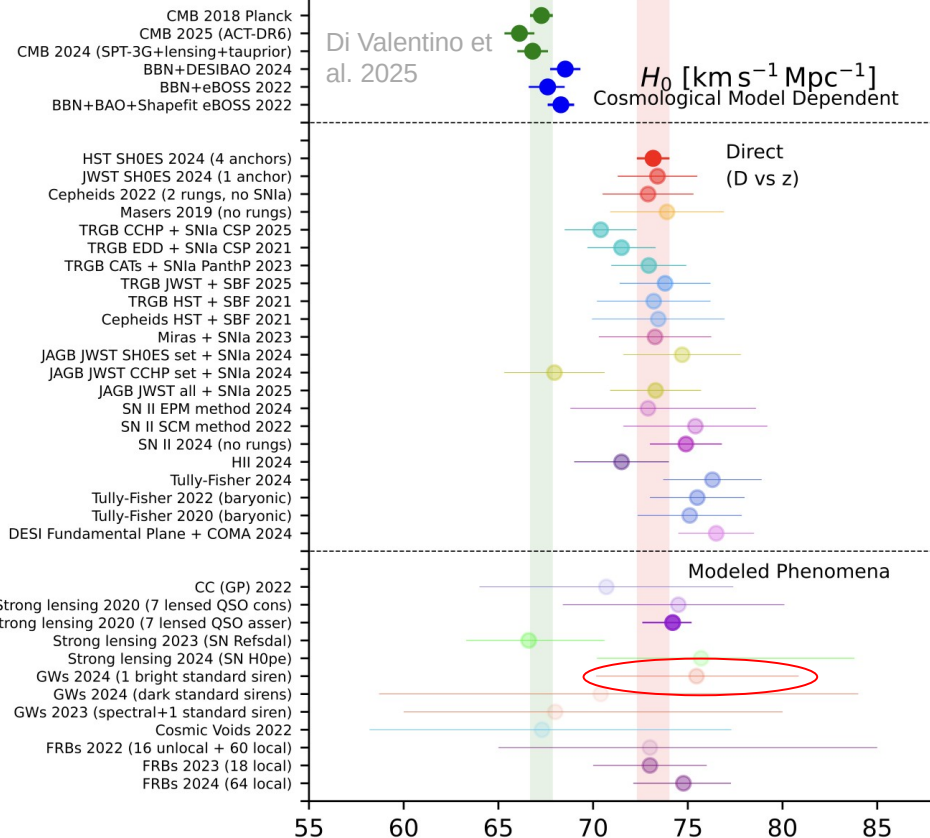
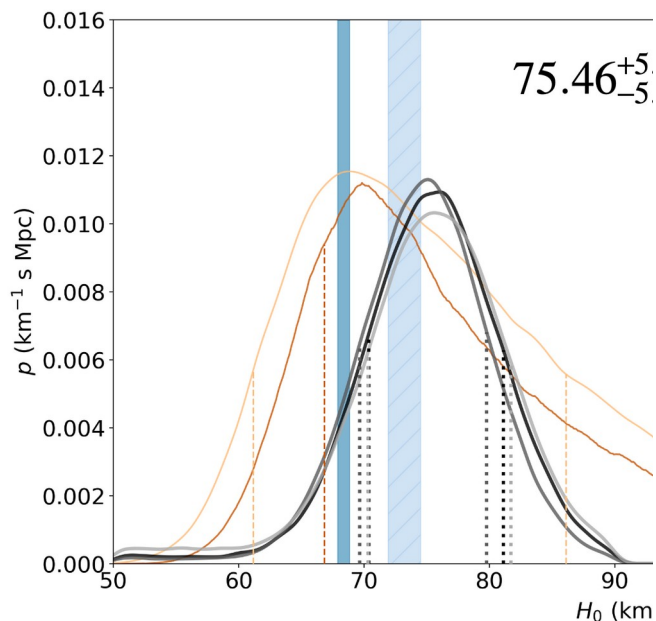


GW170817 bright standard siren

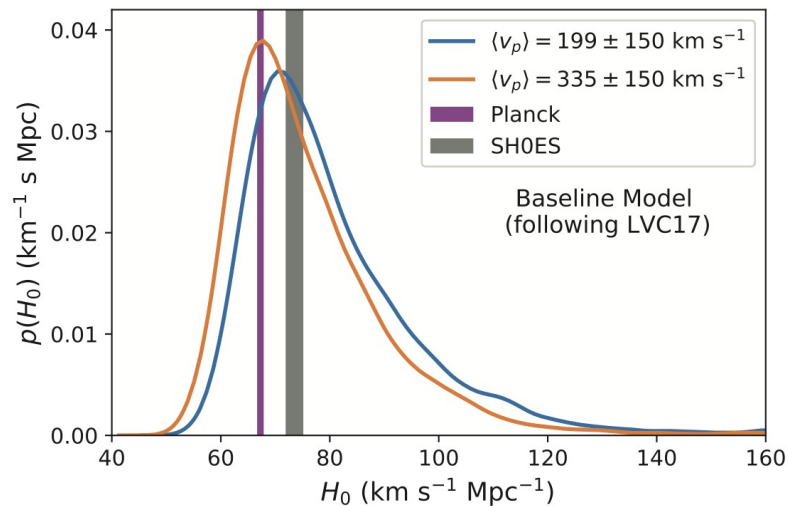
Joint GW-EM inference with host galaxy

Tests of the effect of assumptions on jet i

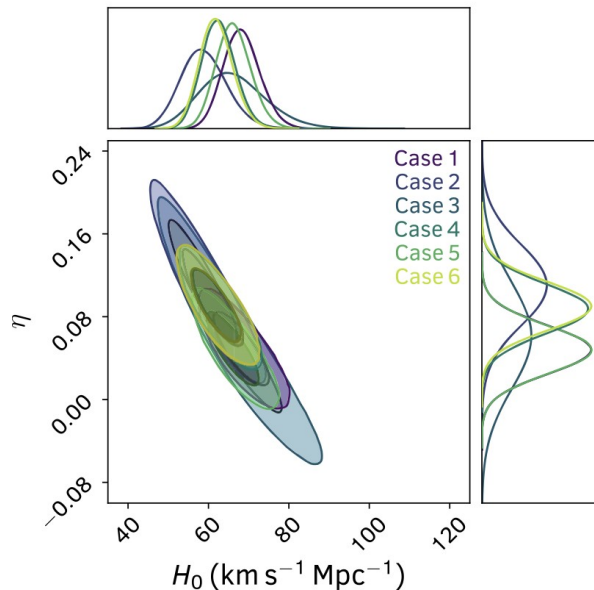
~7% H_0 precision measurement (vs 14%)



GW170817 bright standard siren - peculiar velocities



Nicoulau+ (2020)
Davis (2020)



(Only 1-3 galaxies
in group have
peculiar velocity
measurements)

Howlett &

See also:

Mukherjee+(2020) ³⁵

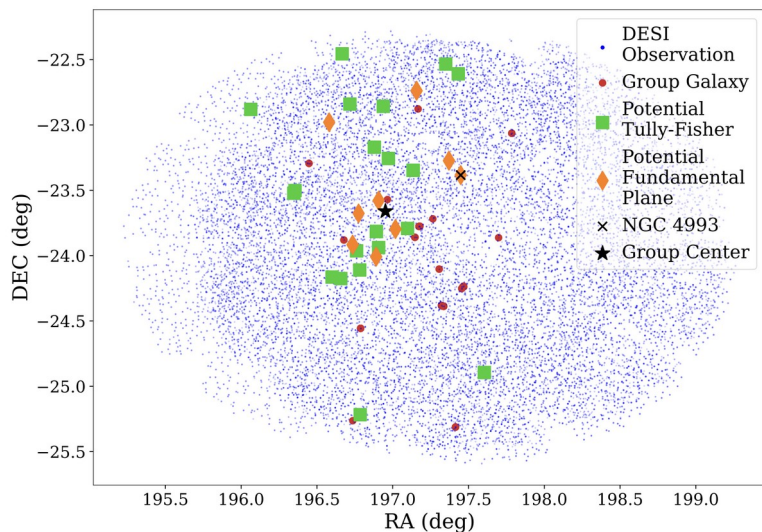
GW170817 as a bright standard siren - DESI

DESI observed 39 group galaxies

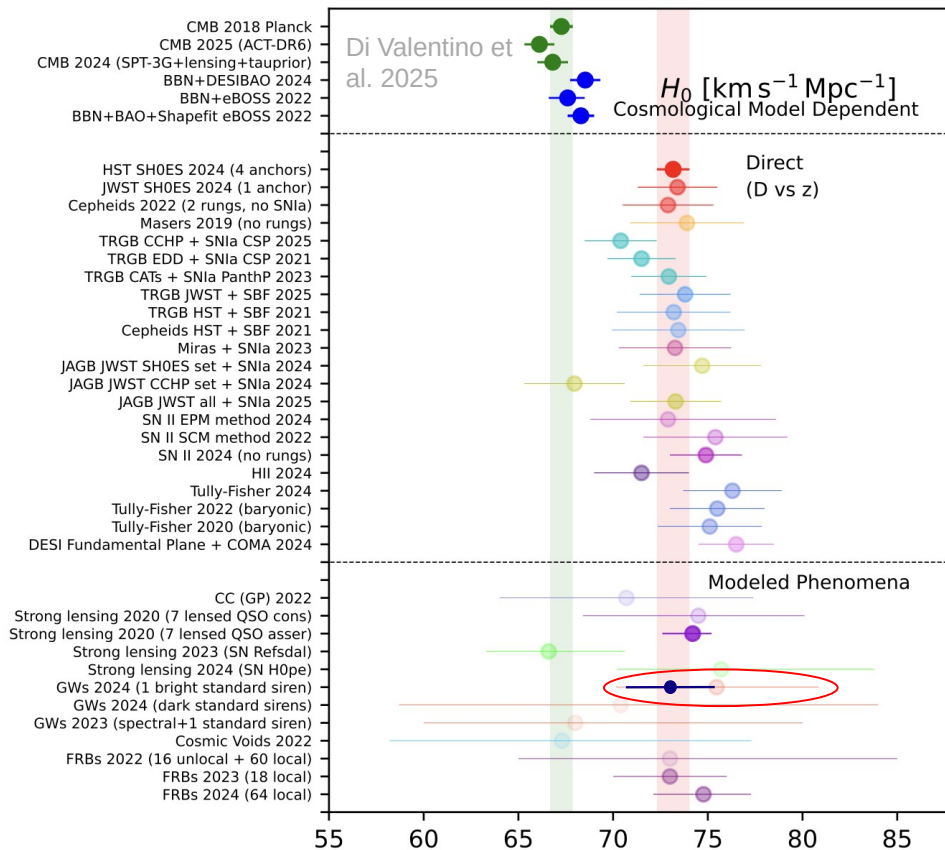
5-fold increase in number of peculiar velocities

$$\text{DESI } H_0 = 70.9^{+6.4}_{-8.5} \text{ km s}^{-1} \text{ Mpc}^{-1}$$

$$\text{Surface Brightness Fluctuations } H_0 = 73.4^{+5.2}_{-5.2}$$



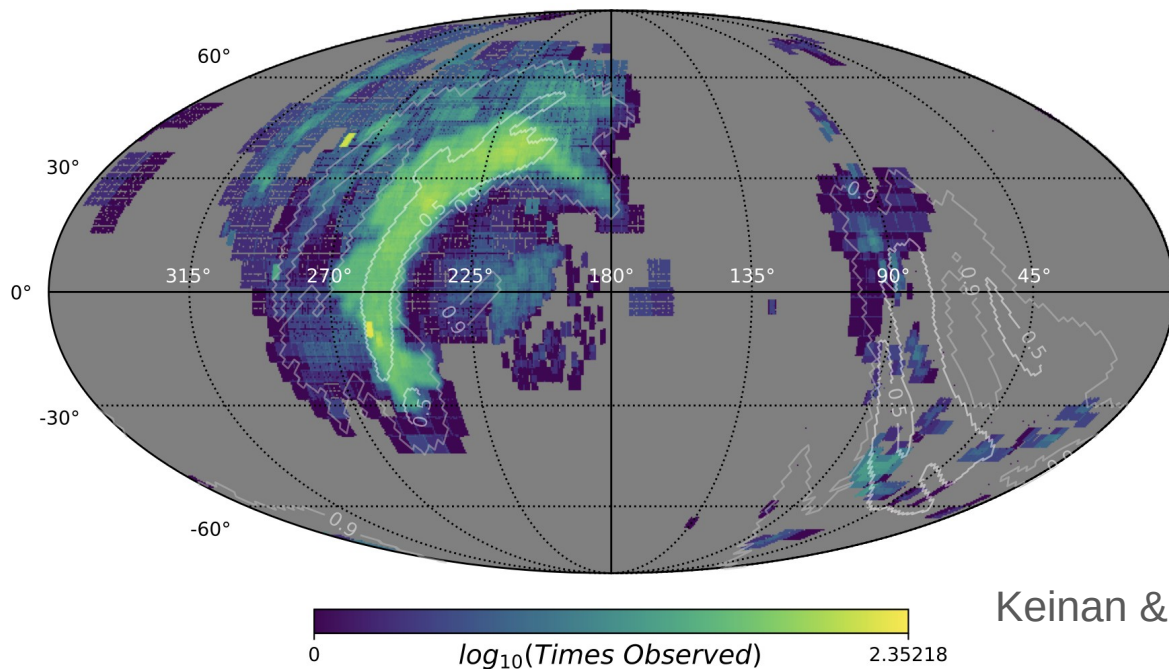
Amsellem, AP, et al. (DESI), 2026, ApJ,



GW190425 - Optical observations

No EM counterpart found (e.g. Hosseinzadeh+2019, Coughlin+2019, Smartt+2024), but:

Single-detector detection (>7000 sq. deg. localization), and ~ 160 Mpc away, massive (small ejecta masses)

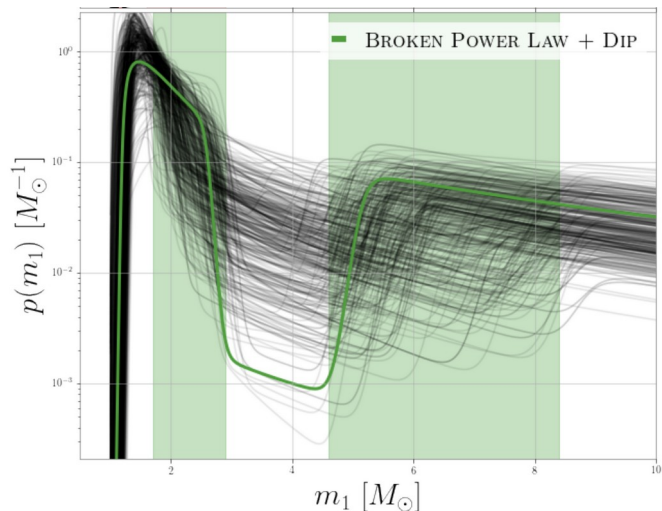


Keinan & Arcavi 2024

Multimessenger KN detection prospects

Simulate BNS+NSBH GW
detections+KNe

Power law+dip+break model



Farah+2022, ApJ; LVK 2024, ApJL

BNS rates updated (rough estimate) based on no
detections of BNS so far in O4

Produce lightcurves to inform follow-up campaigns

Run	Binary type	GW detection rate (yr^{-1})	KN detectability (yr^{-1})				
			Argus		Rubin		
			g	r	g	r	i
O5	BNS	$4.95^{+7.38}_{-3.76}$	$3.22^{+4.81}_{-2.45}$	$3.25^{+4.86}_{-2.48}$	$3.97^{+5.93}_{-3.02}$	$3.25^{+5.96}_{-2.48}$	$3.64^{+5.44}_{-2.77}$
	NSBH	$13.56^{+11.66}_{-7.58}$	$0.12^{+0.10}_{-0.07}$	$0.16^{+0.14}_{-0.09}$	$0.34^{+0.29}_{-0.19}$	$0.40^{+0.34}_{-0.22}$	$0.32^{+0.28}_{-0.18}$

↑
Pretty much all
Mass gap

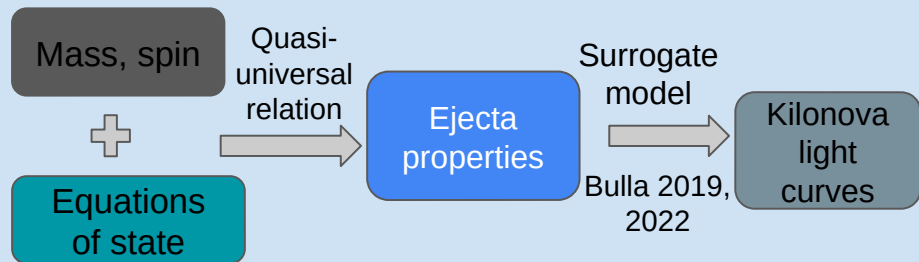
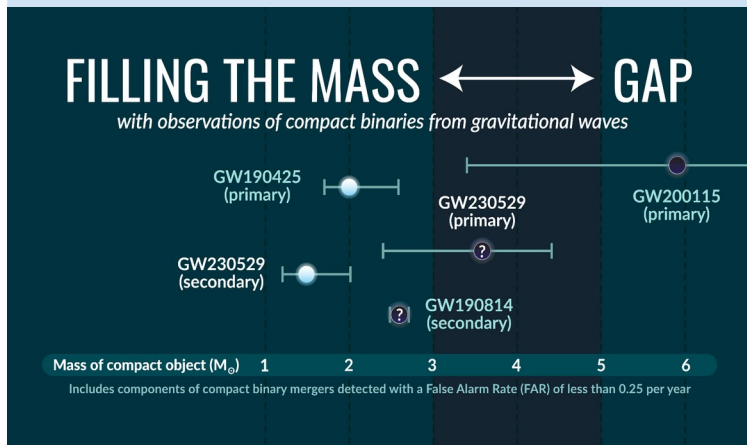


Neutron star- black hole mergers

Kilonovae from mass-gap NSBH mergers

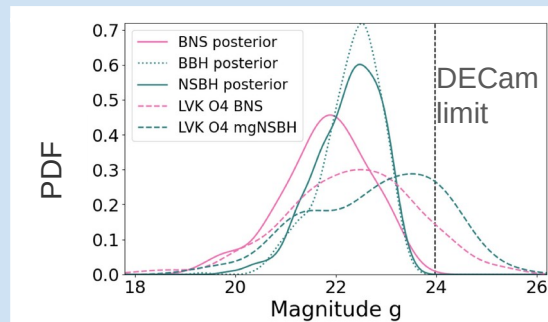
GW230529: likely NSBH merger with 2.5-4.5 $M_{\odot}$ black hole (Abbott et al. 2024, ApJL, arXiv: 2404.04248)

No EM counterpart detected



Kilonova production: 2-41% chance if NSBH

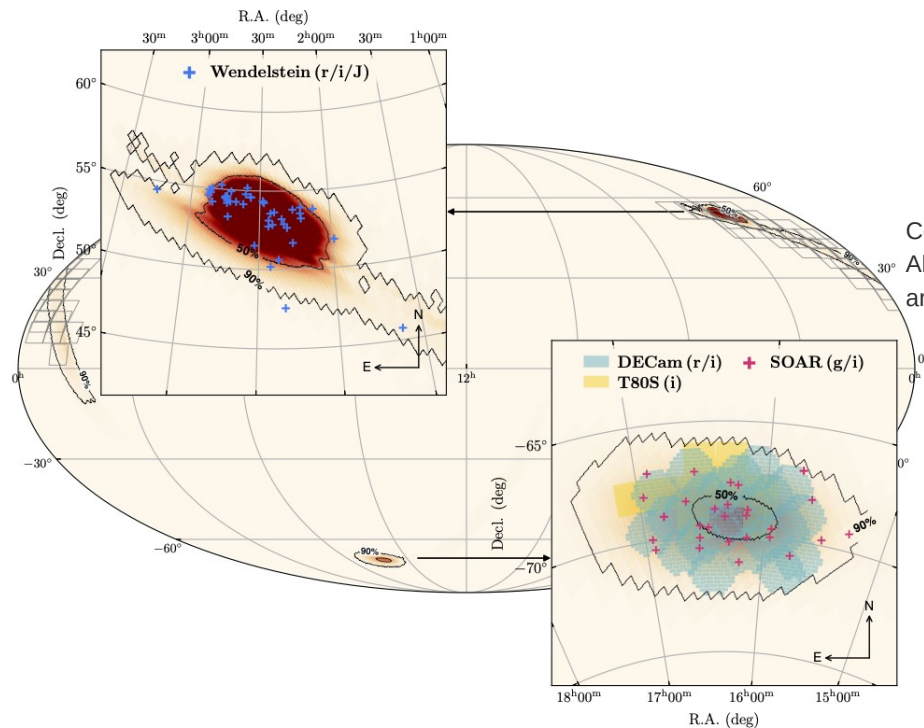
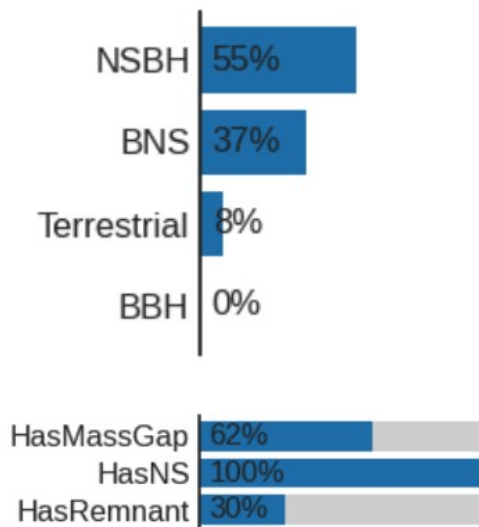
Kilonova detectability: High detectability in optical



Kunnumkai, AP, Bulla, Dietrich, Farah & Pang, 2024, arXiv:2409.10651

S250206dm - Kilonova search

Hu, Cabrera, AP, et al.,
2025 arXiv:2506.2262



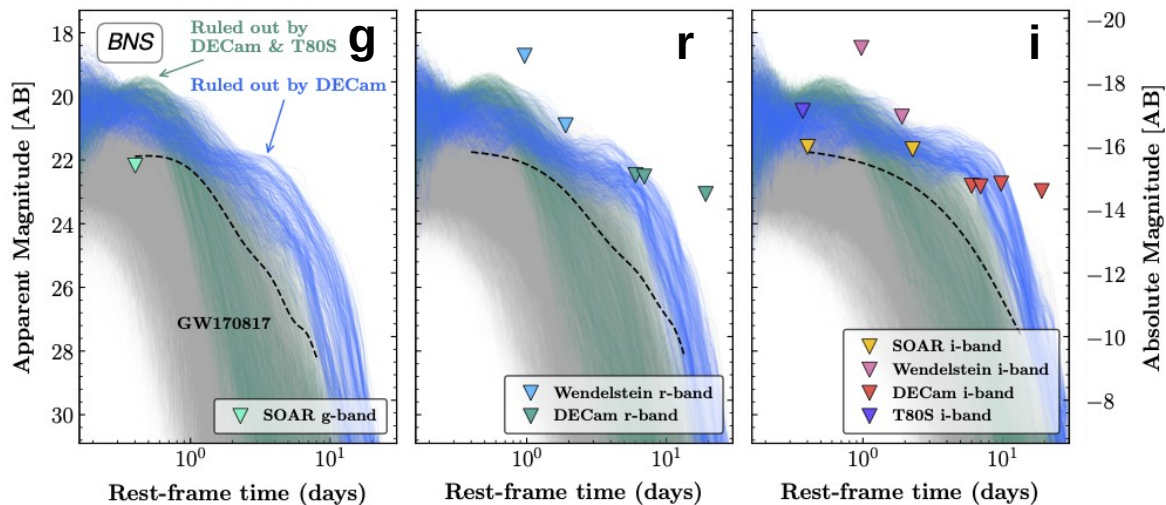
Coordinated with ZTF:
Ahumada et al. 2025,
arXiv:2507.003576

S250206dm - GW-MMADS DECam search

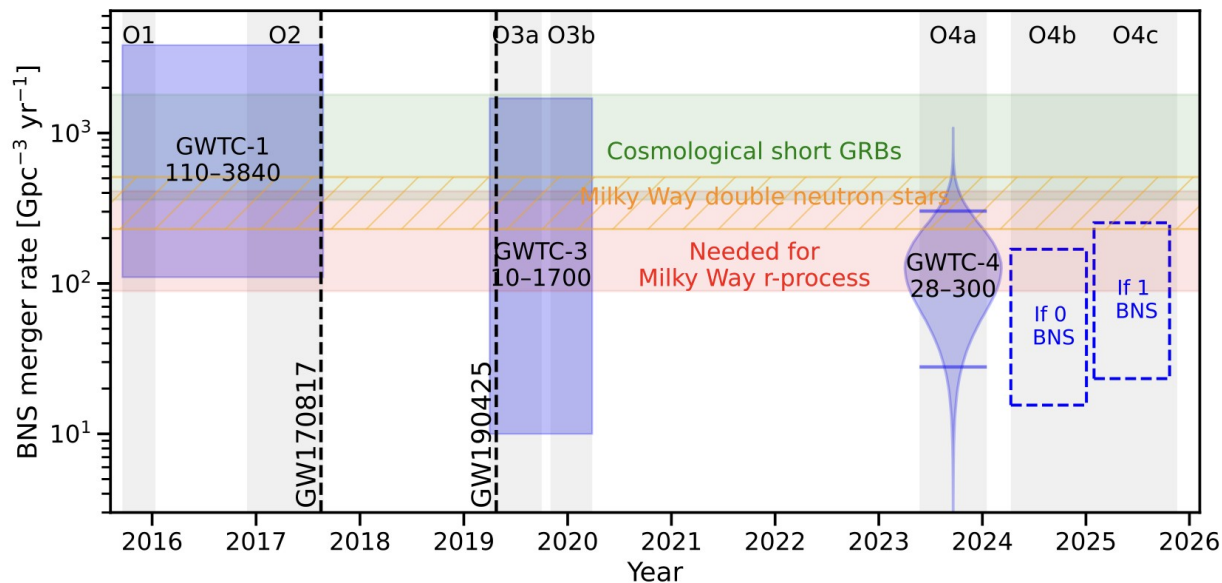
Hu, Cabrera, AP, et al.,
2025 arXiv:2506.2262

Rule out **>40% kilonova models brighter/similar to GW170817** kilonova at **~320 Mpc**

If prompt observations were possible, we would have excluded **>80%** of all models (or found the counterpart!)



Where are the EM-bright neutron star mergers?

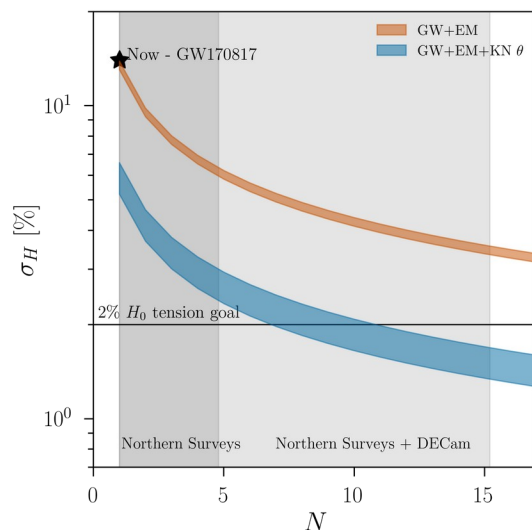


Fishbach et al. 2026

See Kunnumkai, AP, O'Connor, Farah, Magana-Hernandez 2026
arXiv:2604.05046

Bright Standard Sirens - Prospects

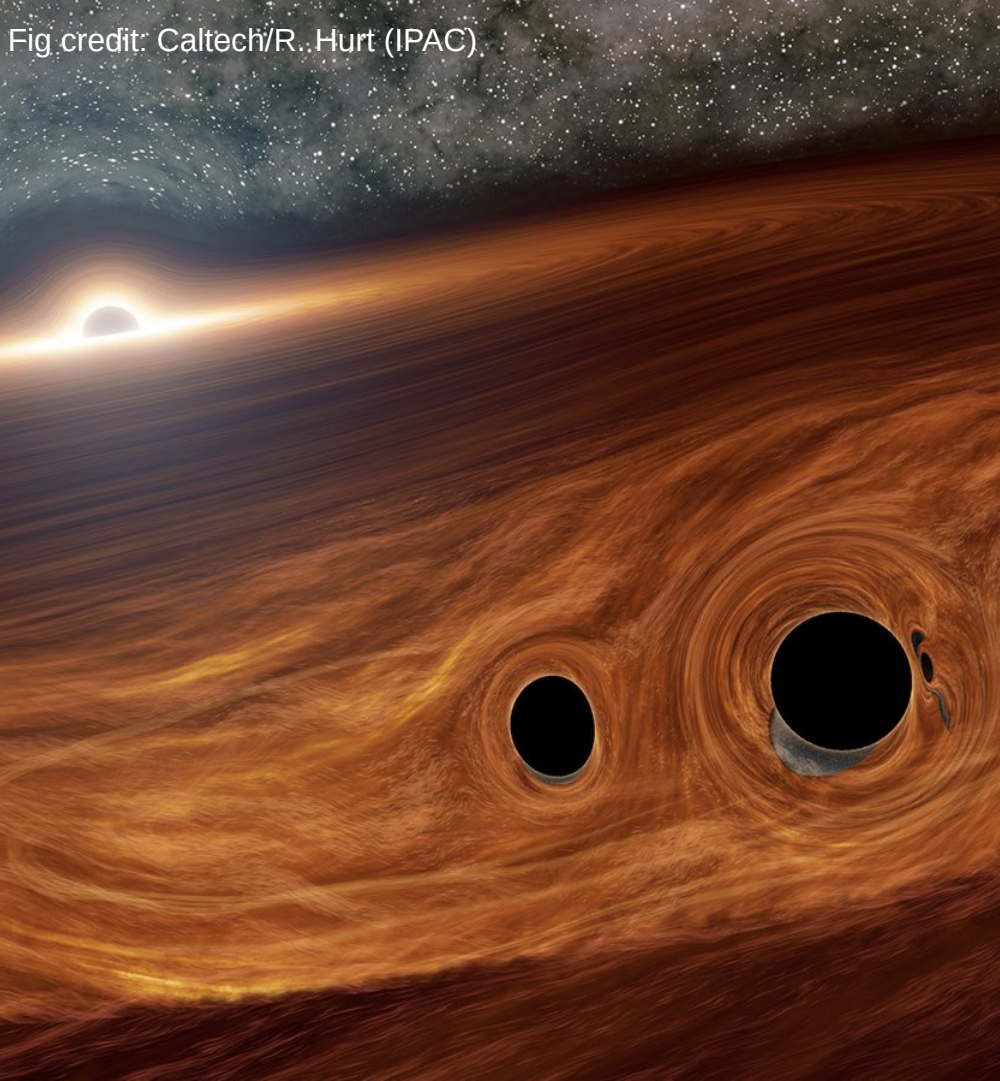
~2% measurement with ~50 events, enough to shed light on H_0 tension
(Chen+2018, Feeney+2019), ~10 with viewing angle constraints
Need to wait for end of O5 (~2031) to reach ~2%



Adapted from AP+2019, arxiv:1903.04730

Potential sources of systematics/limitations:

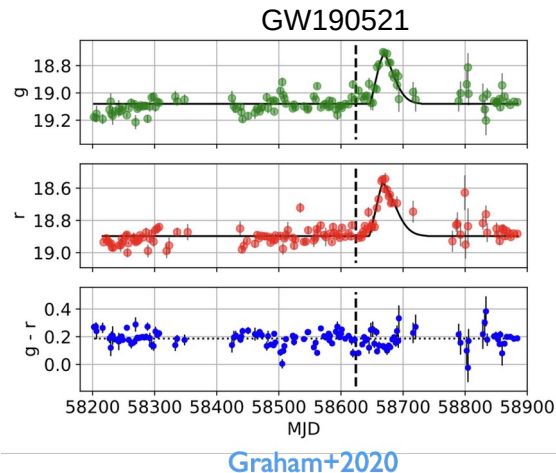
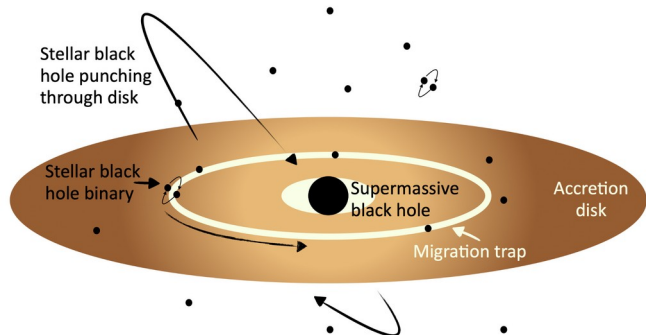
- GW selection effects
- GW modeling (e.g. Kunert+2024)
- Peculiar velocity (nearby sources)
- Jet modeling (e.g. Gianfragna+2024), jet misalignment (Muller+2024)
- EM selection effects (e.g. Chen+ 2024, Mancarella+2024)



Binary black hole mergers

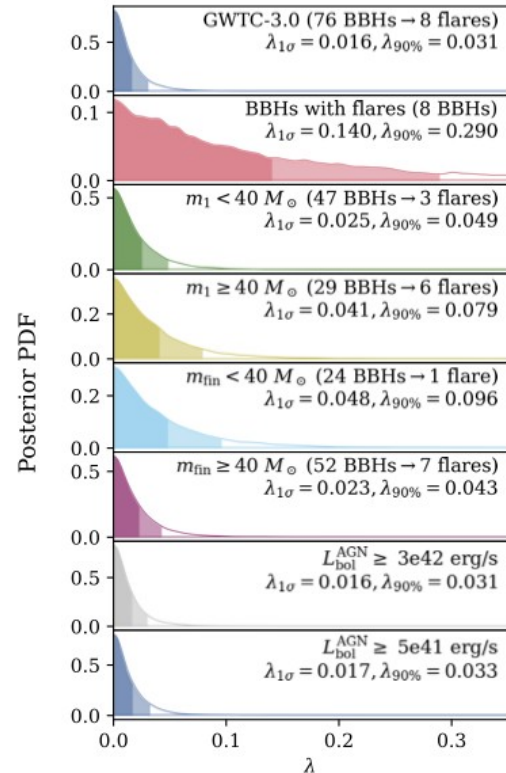
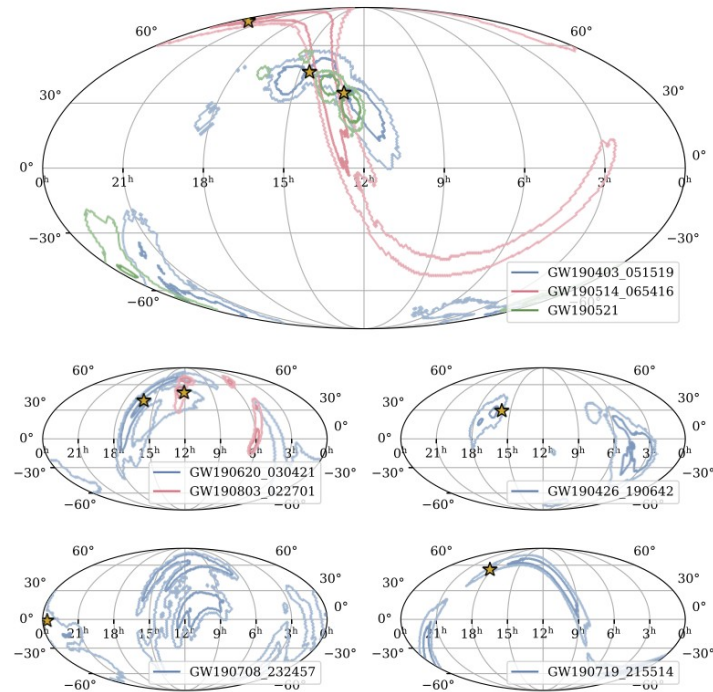
Do LIGO/Virgo/KAGRA BBH produce AGN flares?

- Ideal environment for formation of massive stellar black holes through repeated mergers of smaller BH in AGN disks
- 7 AGN flare in coincidence with GW190521 (ZTF, Graham+2020, PRL arXiv:2006.14122), and other events (Graham et al 2023, ApJ, arXiv:2209.13004)



GWTC-3 multimessenger constraints

- ZTF Flares from Graham et al. (2023) + GWTC-3
- **<3% of LVK BBH mergers give rise to ZTF-observable AGN flares (90%CI)**
- Up to **40% of LVK BBH can still originate in AGN disks**, consistent with theoretical predictions
- Fainter AGNs/flares to be probed

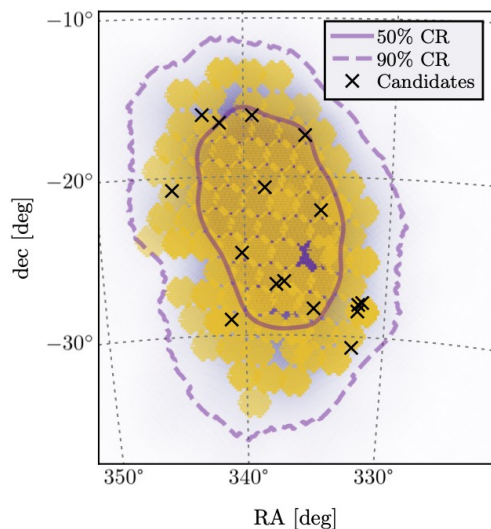


Binary black hole with $M_{\text{tot}} \sim 70 M_{\odot}$

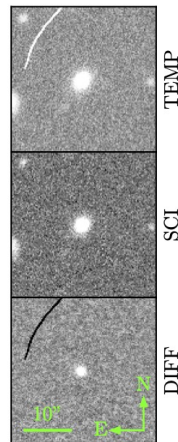
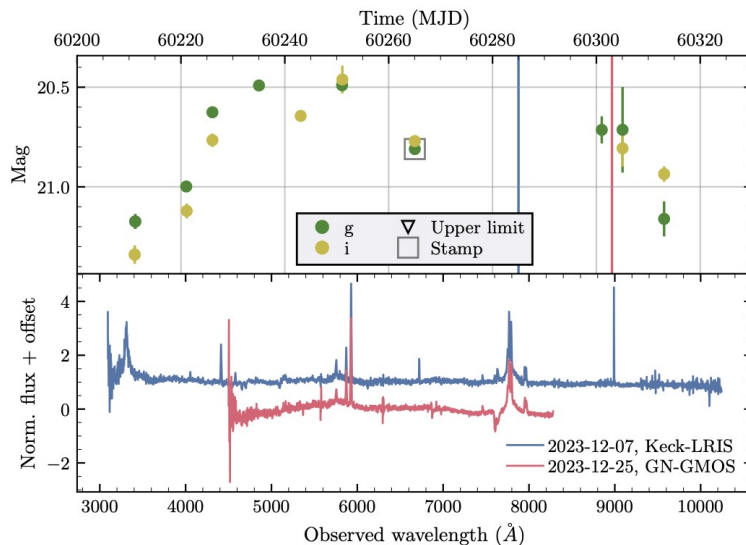
AT 2023aagj: DECam nuclear transient in AGN, $>5\sigma$ deviation from typical AGN variability

Spectral variability indicative of off-center source (predicted in McKernan et al. 2019), Bondi accretion interpretation

First *live* search and multiwavelength, photometric, and spectroscopic monitoring of a GW **BBH** optical

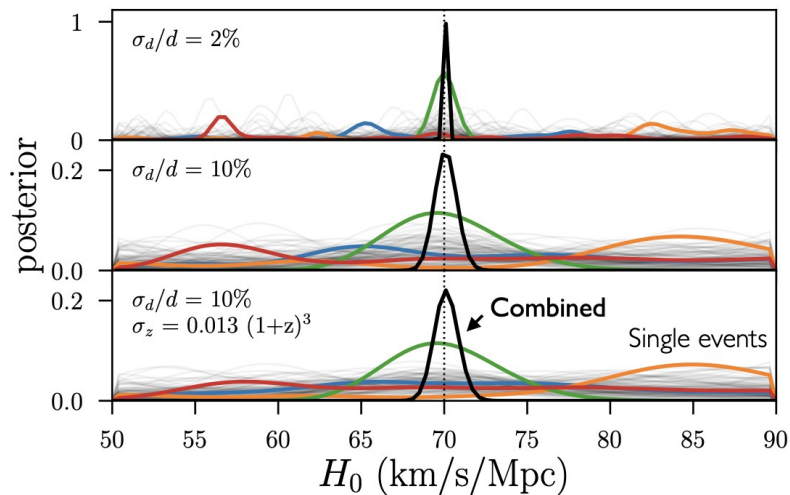


he



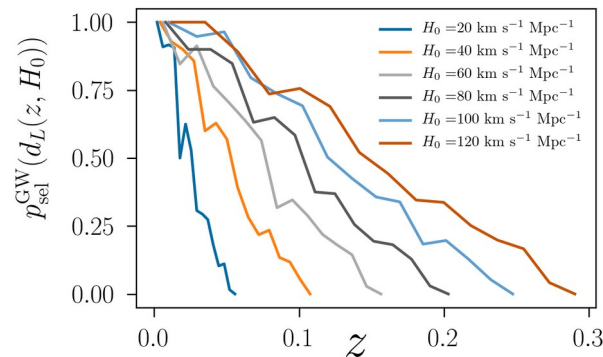
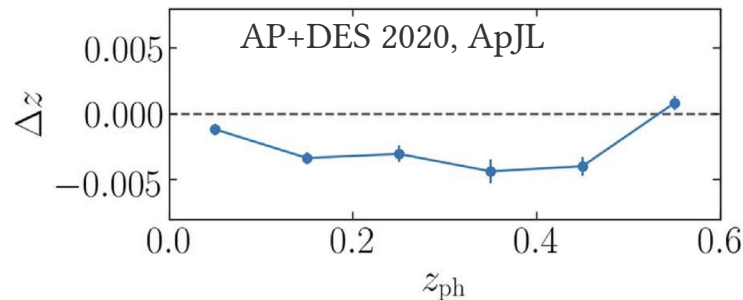
Dark sirens - simulations

- Dark Energy Survey-like galaxy simulations (DeRose et al. 2019, Wechsler et al. 2021)
- Simulated GW events on a galaxy at z with $H_0=70$ km/s/Mpc
- **Single events:** posterior expected to have peaks corresponding to large scale structure along the line of sight
- **Peaks are broadened and blended** if d or z uncertainty increases.
- **Converge to the input value** of H_0 from combining enough events ($O(100)$)



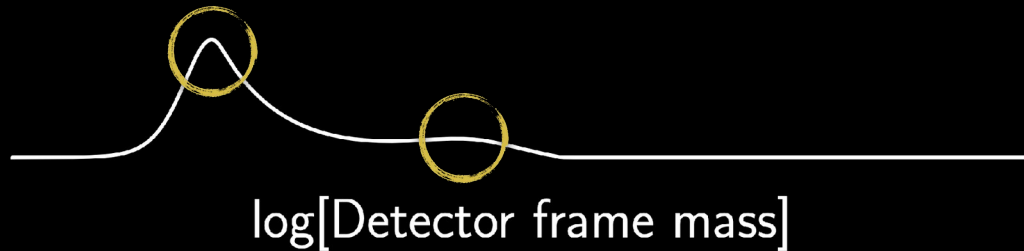
Dark sirens systematics

- Systematics related to **photometric redshifts**: bias correction, full redshift PDF
- **GW Selection effects**
- **Galaxy catalog completeness** corrections (e.g. Fishbach+2019, Gray+2020)
- Expectations from **different BBH formation scenarios**
 - Some work on galaxy **weighing** Perna et al. 2024; Hanselman et al. 2025)

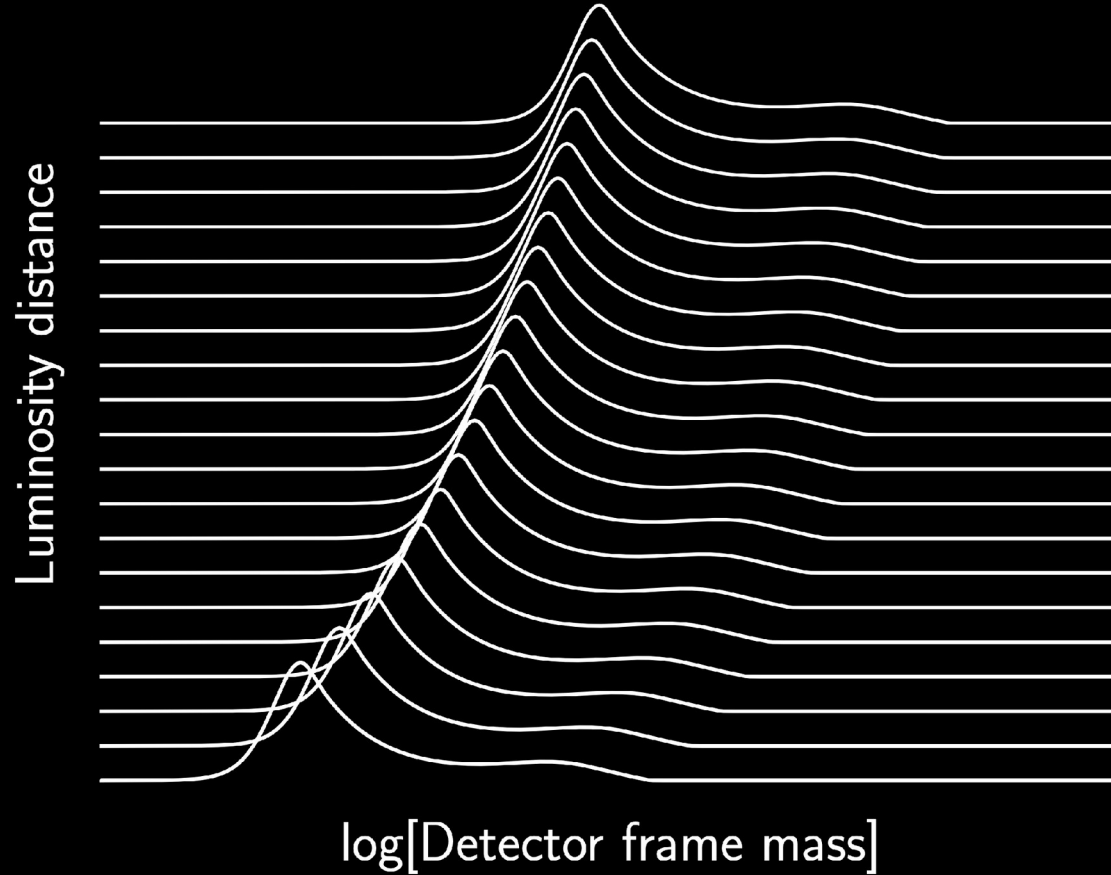


SPECTRAL SIRENS

$$\{d_L(z), m_{\text{det}} = (1+z)m\}$$



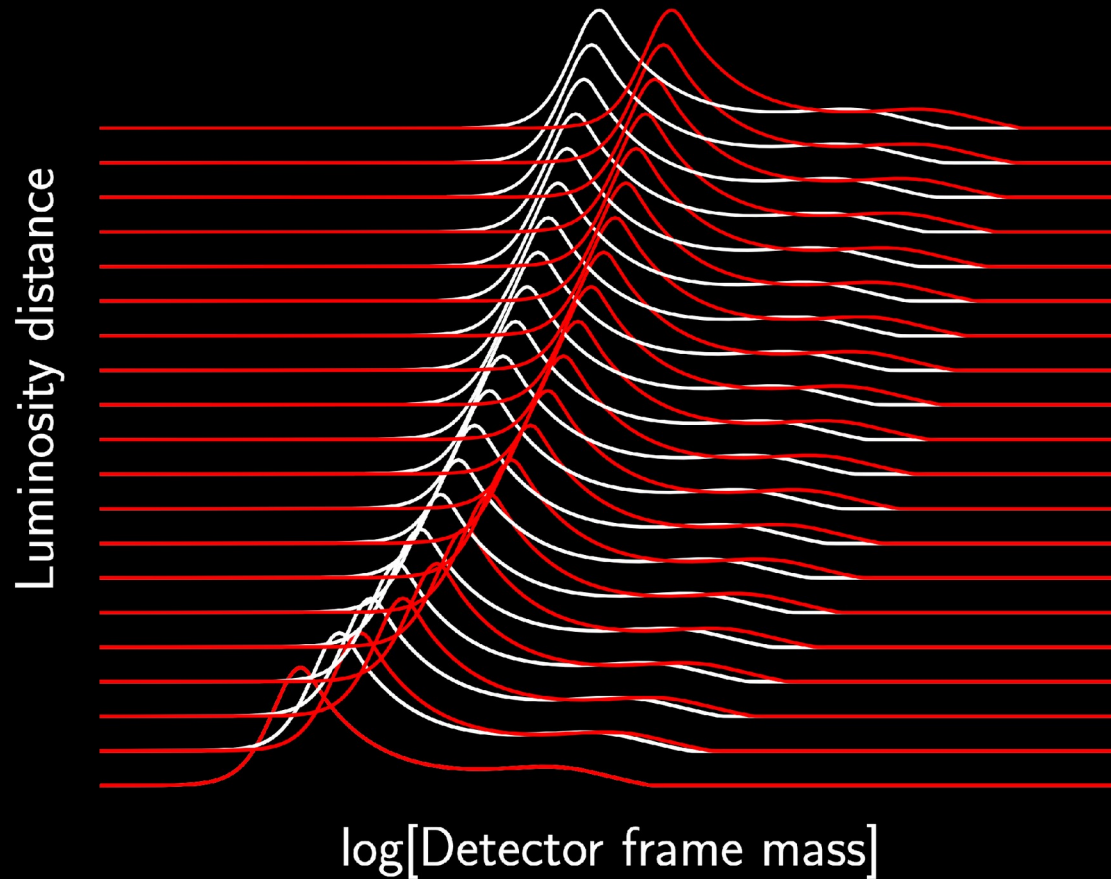
SPECTRAL SIRENS

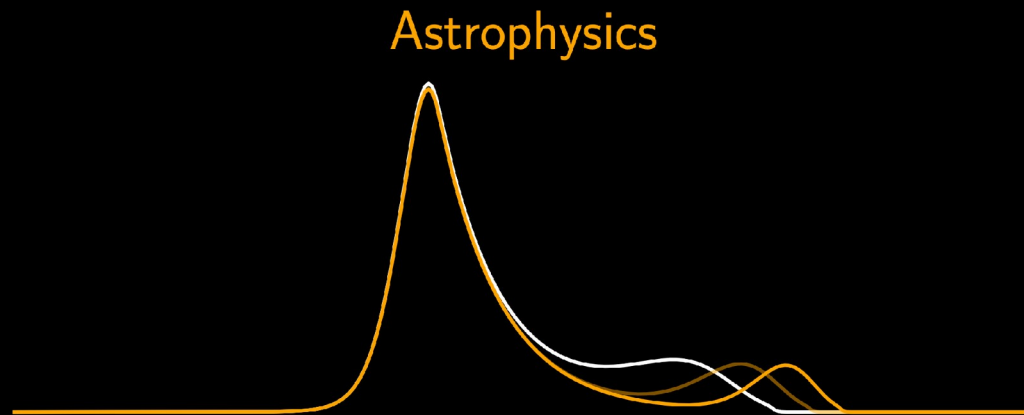
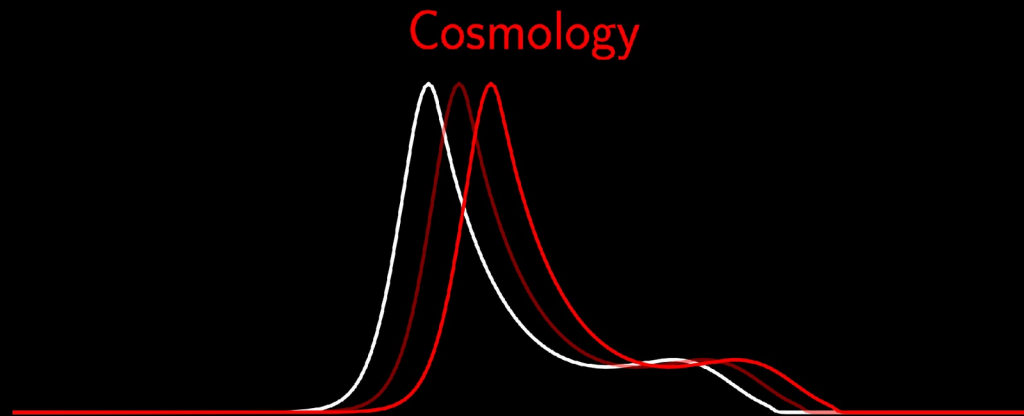


Credit: J. Ezquiaga

SPECTRAL SIRENS

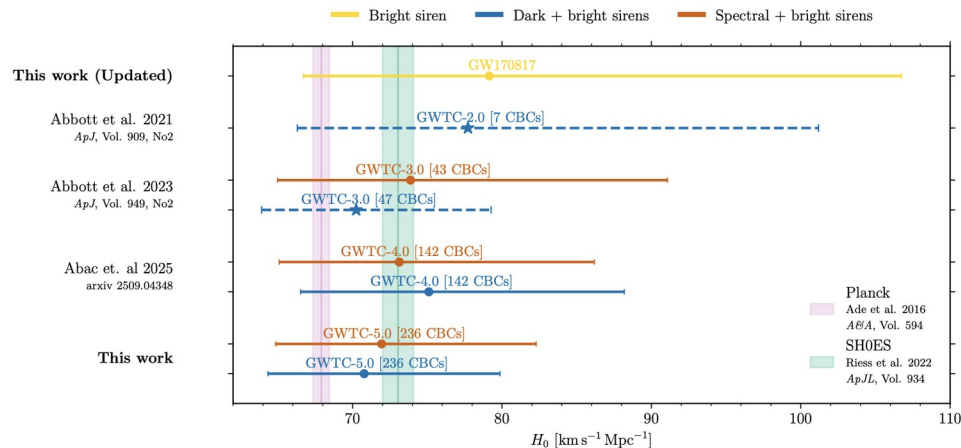
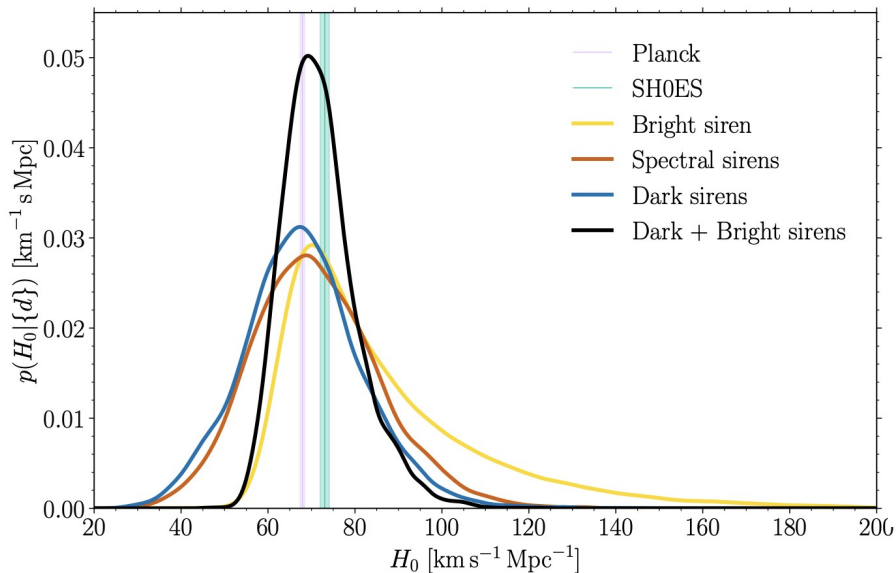
higher H_0





$\log[\text{Detector frame mass}]$

Dark and spectral siren measurements with GWTC-5

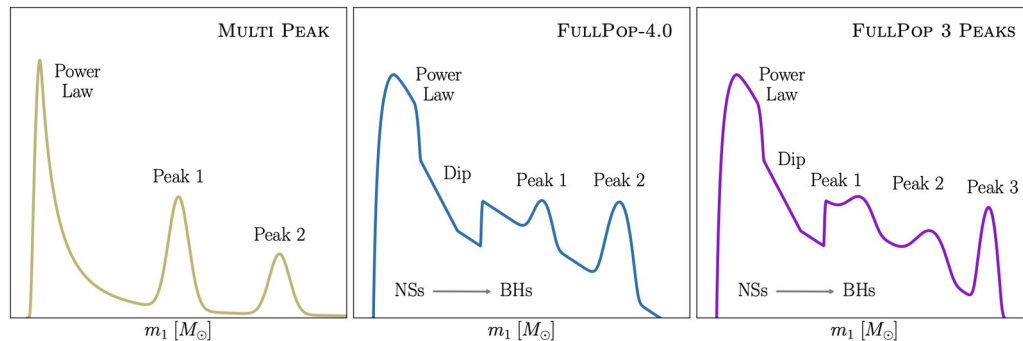
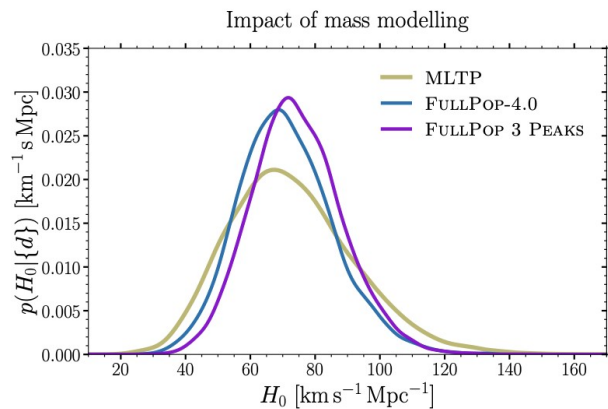


GLADE+ and DES galaxy catalogs

Marginalize over possible population

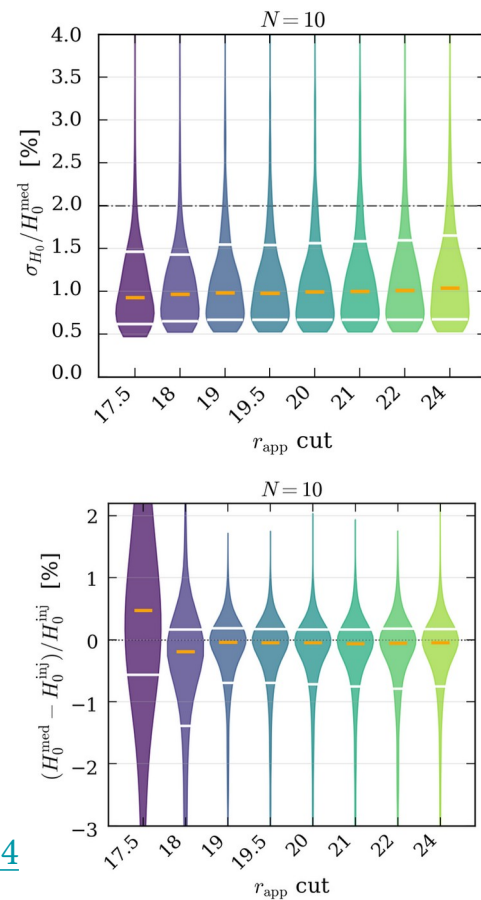
~26% precision improvement compared to GWTC-4 (up to O4a)

Dark and spectral siren measurements with GWTC-5



Dark siren prospects for current generation detectors

- Realistic O5 and A# black hole mergers simulation+Uchuu LSST-like cosmological simulation
- Top 10 A# events/year $\rightarrow$ 1-1.5% H_0 precision (achievable ~2030s?)
- Top 10 O5 events $\rightarrow$ 1-4% precision in O5
- DESI-like volume limited sample provides unbiased results
- Minimal impact of weighting scheme

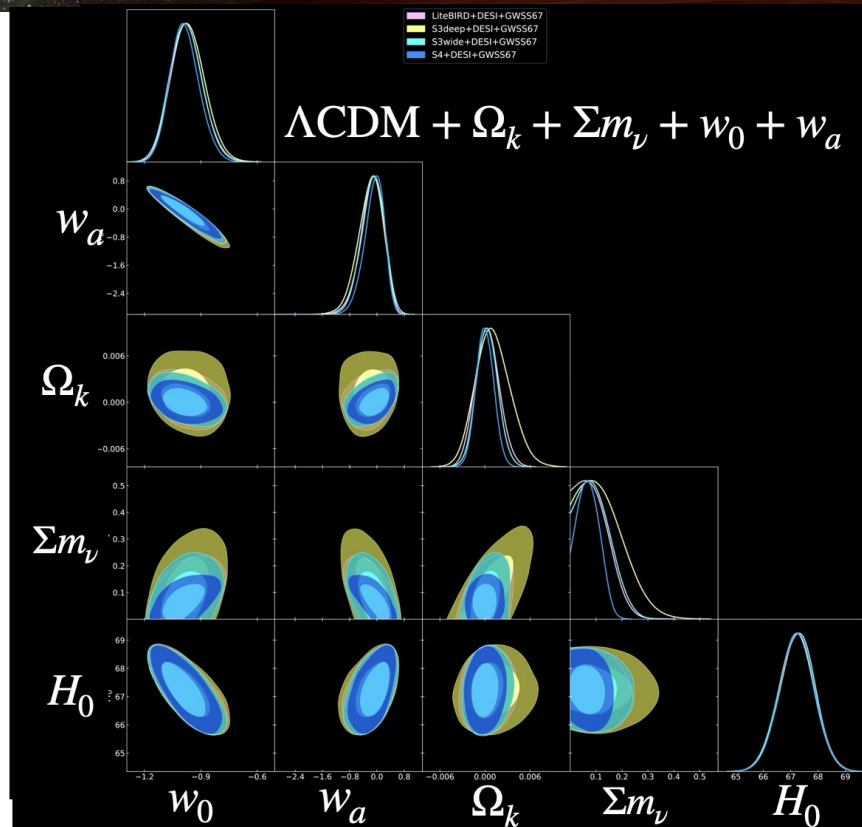


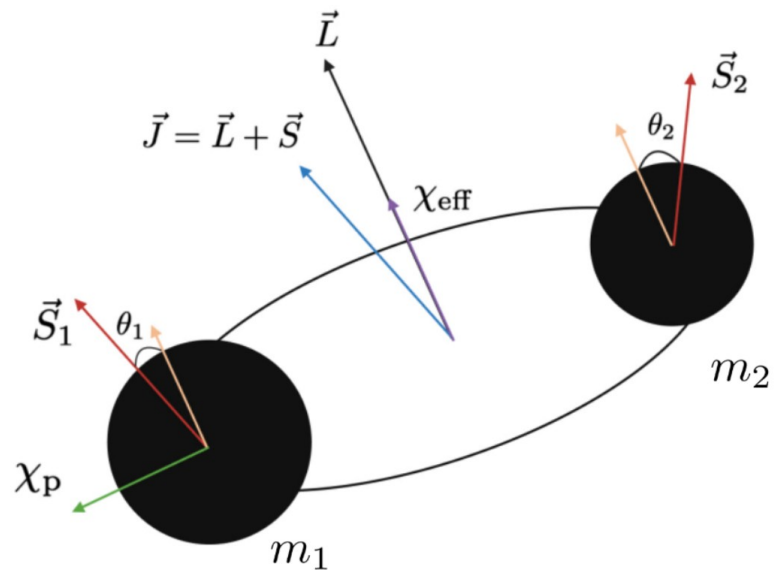
Conclusions

- **Standard sirens** are unique cosmological probes
- **2% H_0 precision needed** to provide insights on **Hubble tension**
 - ~10 bright standard sirens with EM viewing angle constraints
 - ~10 extremely well localized dark standard sirens
 - Possible by ~2032
- Standard sirens can measure **cosmological parameters beyond H_0**
- Different kinds of standard sirens are **complementary** and powerful when combined
- There is **a lot more** to learn from combining GW+EM observations: **modified gravity, neutron star equation of state and nuclear physics, origin of black holes and neutron stars, the formation of the heavy elements** in the Universe

Bright sirens impact on dark energy parameters

- Combining upcoming GW H_0 constraints (1% precision) + future CMB + BAO significantly **improves constraints beyond Λ CDM**
- Breaks geometrical degeneracies** between parameters from CMB
- Factor 1.6-2.8 improvement on dynamical DE parameters





Standard siren inference

Hierarchical Bayesian inference in presence of selection effects and measurement uncertainties - modified Poisson process (Mandel+2019)

Likelihood of obtaining N_{obs} observations, each described by some parameters θ , in a GW data collection $\{x\}$, for non overlapping signals, for a given observing time T_{obs} :

$$\begin{aligned}\mathcal{L}(\{x\}|\Lambda) &\propto e^{-N_{\text{exp}}(\Lambda)} \prod_{i=1}^{N_{\text{obs}}} \int \mathcal{L}(x_i|\theta) \frac{dN}{dtd\theta}(\Lambda) dt d\theta \\ &\propto e^{-N_{\text{exp}}(\Lambda)} \prod_{i=1}^{N_{\text{obs}}} T_{\text{obs}} \int \mathcal{L}(x_i|\theta) \frac{dN}{dtd\theta}(\Lambda) d\theta.\end{aligned}\quad \mathcal{L}(x|\Lambda) \propto \prod_{i=1}^{N_{\text{obs}}} \frac{\int \mathcal{L}(x_i|\theta) p_{\text{pop}}(\theta|\Lambda) d\theta}{\int p_{\text{det}}(\theta) p_{\text{pop}}(\theta|\Lambda) d\theta}.$$

Focus on parameters of interest and move to source frame with a cosmology:

$$\mathcal{L}(x|\Lambda) \propto \prod_{i=1}^{N_{\text{obs}}} \frac{\int \mathcal{L}(x_i|z, \mathbf{m}, \cos \iota, \mathbf{\Omega}, H_0) p_{\text{pop}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}|\Lambda) dz d\mathbf{m} d\cos \iota d\mathbf{\Omega}}{\int p_{\text{det}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}, H_0) p_{\text{pop}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}|\Lambda) dz d\mathbf{m} d\cos \iota d\mathbf{\Omega}}.$$

Explicitly model: the population distribution of CBCs in the source frame

$$p_{\text{pop}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}|\Lambda) = p_{\text{pop}}(z|\Lambda) p_{\text{pop}}(\mathbf{m}|\Lambda) p_{\text{pop}}(\cos \iota|\Lambda) p_{\text{pop}}(\mathbf{\Omega}|\Lambda).$$

Standard siren inference - galaxy catalog (dark sirens)

Defining a population distribution conditioned on galaxy catalog CAT:

$$p_{\text{pop}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega} | \mathbf{\Lambda}, \text{CAT}) = p_{\text{pop}}(z, \mathbf{\Omega} | \text{CAT}, \mathbf{\Lambda}) p_{\text{pop}}(\mathbf{m} | \mathbf{\Lambda}) p_{\text{pop}}(\cos \iota | \mathbf{\Lambda}).$$

$$p_{\text{pop}}(z, \mathbf{\Omega} | \text{CAT}, \mathbf{\Lambda}) \propto \int \delta(z - z_g) p_{\text{gal}}(z_g, \mathbf{\Omega}, \zeta | \text{CAT}) p_{\text{host}}(\zeta, z_g | \mathbf{\Lambda}) d\zeta dz_g.$$

$$p_{\text{pop}}(z, \mathbf{\Omega} | \text{CAT}, \mathbf{\Lambda}) = f(z, \mathbf{\Omega}) p_{\text{pop}}(z, \mathbf{\Omega} | \text{CAT}, \mathbf{\Lambda}) + [1 - f(z, \mathbf{\Omega})] p_{\text{pop}}(z | \mathbf{\Lambda}) p_{\text{pop}}(\mathbf{\Omega} | \mathbf{\Lambda}),$$

$$f(z, \mathbf{\Omega}) = \frac{\int_{M_{\text{thr}}(z, \mathbf{\Omega})}^{\infty} \text{Sch}(M, z) p_{\text{host}}(M, z | \mathbf{\Lambda}) dM}{\int_{M_{\text{faint}}}^{\infty} \text{Sch}(M, z) p_{\text{host}}(M, z | \mathbf{\Lambda}) dM},$$

Standard siren inference - bright sirens

$$\mathcal{L}(x|\Lambda) \propto \prod_{i=1}^{N_{\text{obs}}} \frac{\int \mathcal{L}(x_i|z, \mathbf{m}, \cos \iota, \mathbf{\Omega}, H_0) \mathcal{L}_{\text{EM}}(y_i|z, \cos \iota, \mathbf{\Omega}) p_{\text{pop}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}|\Lambda) dz d\mathbf{m} d\cos \iota d\mathbf{\Omega}}{\int p_{\text{det}}^{\text{GW+EM}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}, H_0) p_{\text{pop}}(z, \mathbf{m}, \cos \iota, \mathbf{\Omega}|\Lambda) dz d\mathbf{m} d\cos \iota d\mathbf{\Omega}}.$$

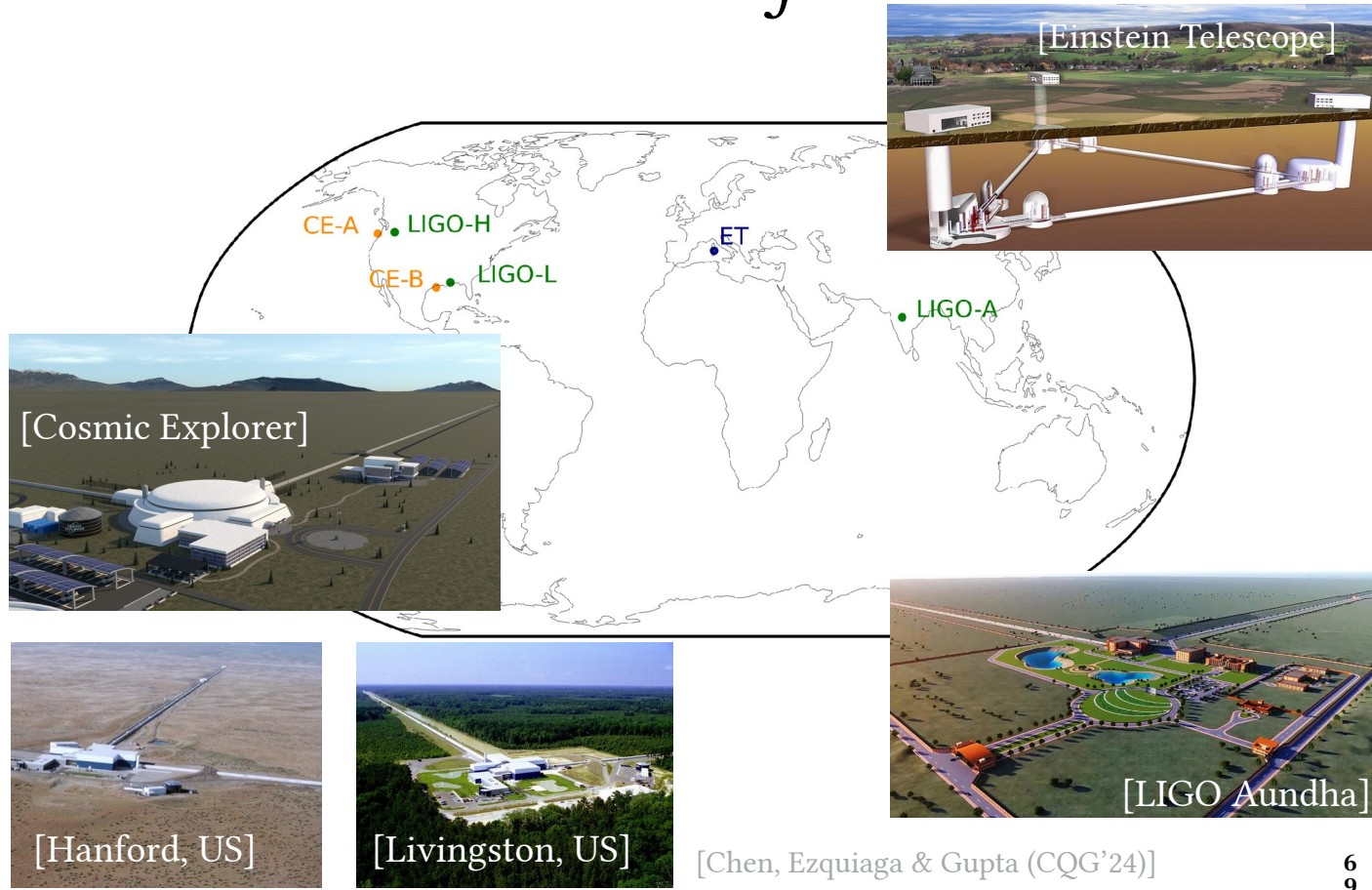
Eq. can simplify:

the redshift is perfectly known, the source frame mass distribution will not introduce any reference

Assume most limiting selection for detection is given by GW detectors

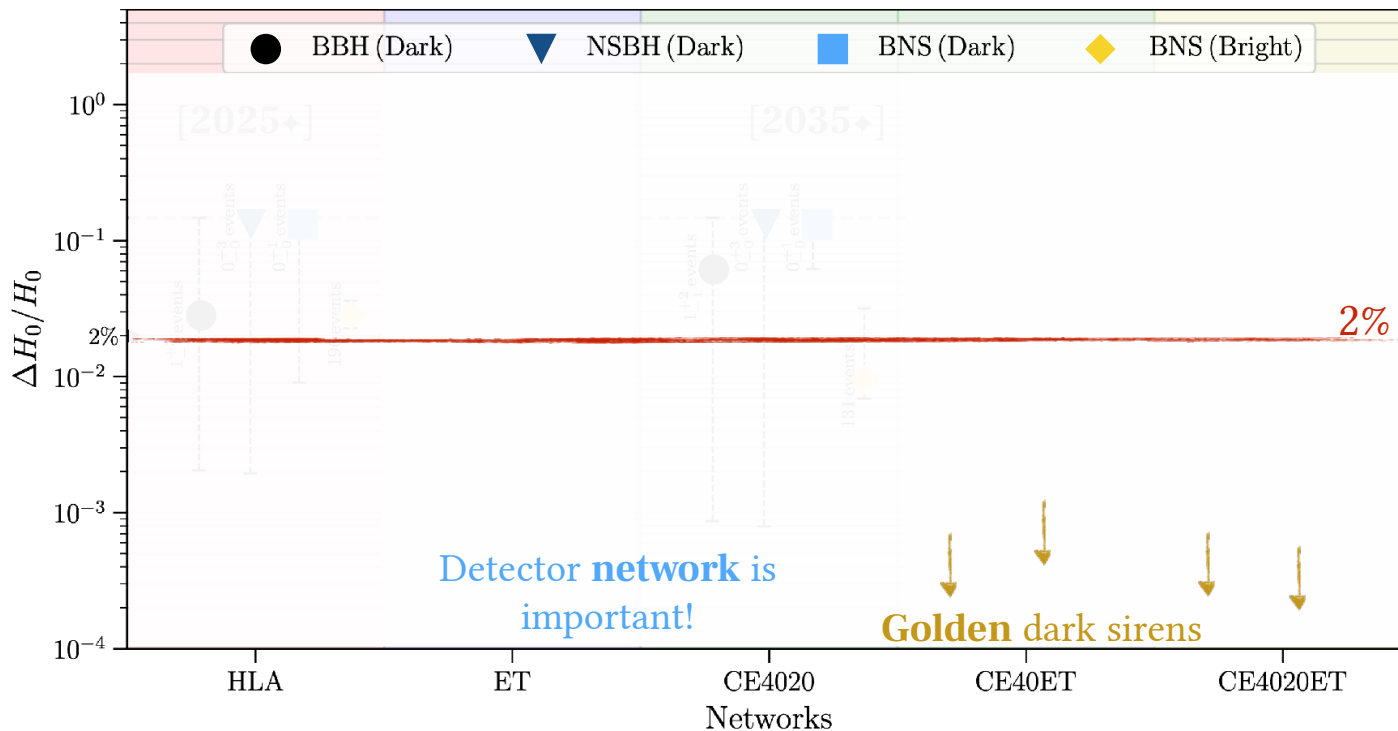
$$\mathcal{L}(x|\Lambda) \propto \prod_{i=1}^{N_{\text{obs}}} \frac{\mathcal{L}(x_i|d_L(z_s, H_0))}{H_0^3}.$$

Standard sirens: *forecasts*



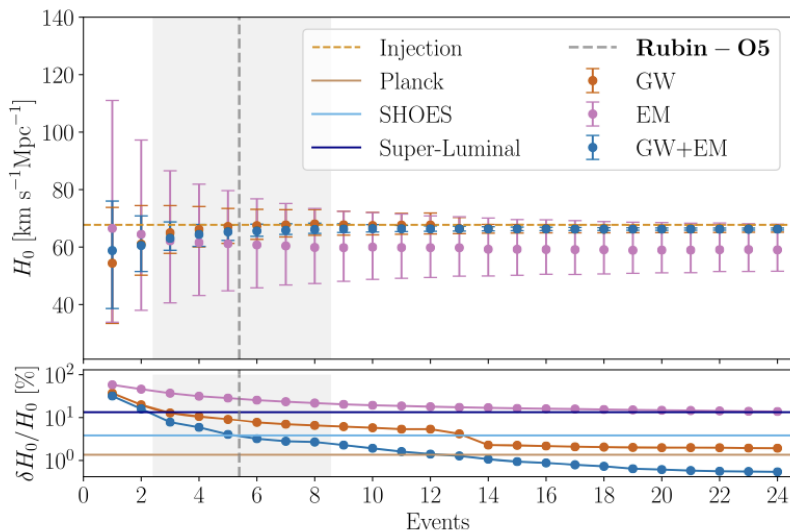
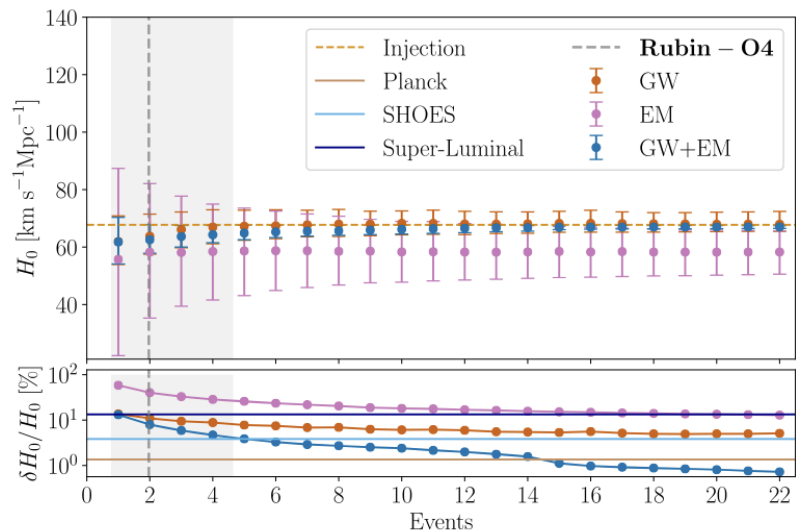
Bright and dark sirens H_0

H: Hanford (US)
L: Livingston (US)
A: Aundha (India)
ET: Einstein Telescope (EU)
CE: Cosmic Explorer (US)



Standard Sirens

Kiendrebeogo et al 2023

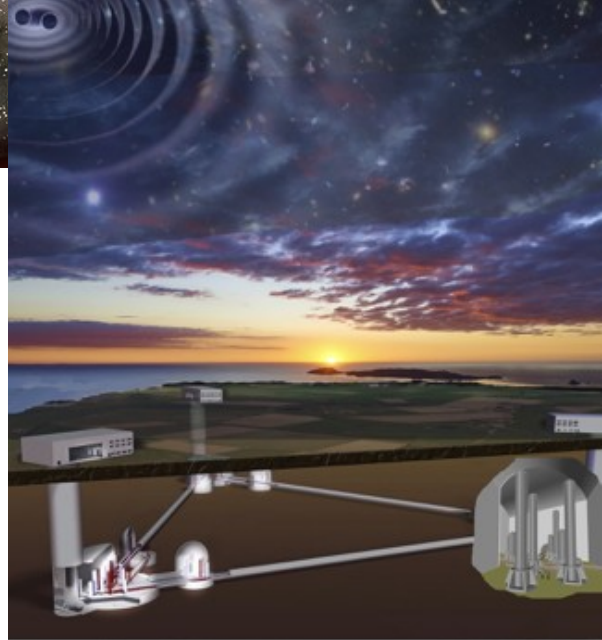


Cosmic Explorer/Einstein Telescope (2030s)

- Huge increase in detection rates:
 - 10^5 - 10^6 BNS per year
- Much higher SNR for mergers in 2G band
- Many mergers (~ 10 k/year) with 1 sq. deg. localization

Combination of CE+ET (or 2+ detectors configurations) improves:

- localization
- distance uncertainty
- early warnings



Peculiar velocities (PV)

- Galaxies' Peculiar Velocities: **motions on top of cosmological expansion**
- Follow inhomogeneous clustering of structures and the laws of gravity
- PV field probes large scale structure and its growth, and gravity
- One possibility is to use galaxies with **distance measurement** (e.g. FP, SNe Ia)
- Use GW BNS from 3G GW detectors - forecast
- We use PV power spectrum, overdensities power spectrum, and their cross-correlation

Power Spectra depend on: Growth index

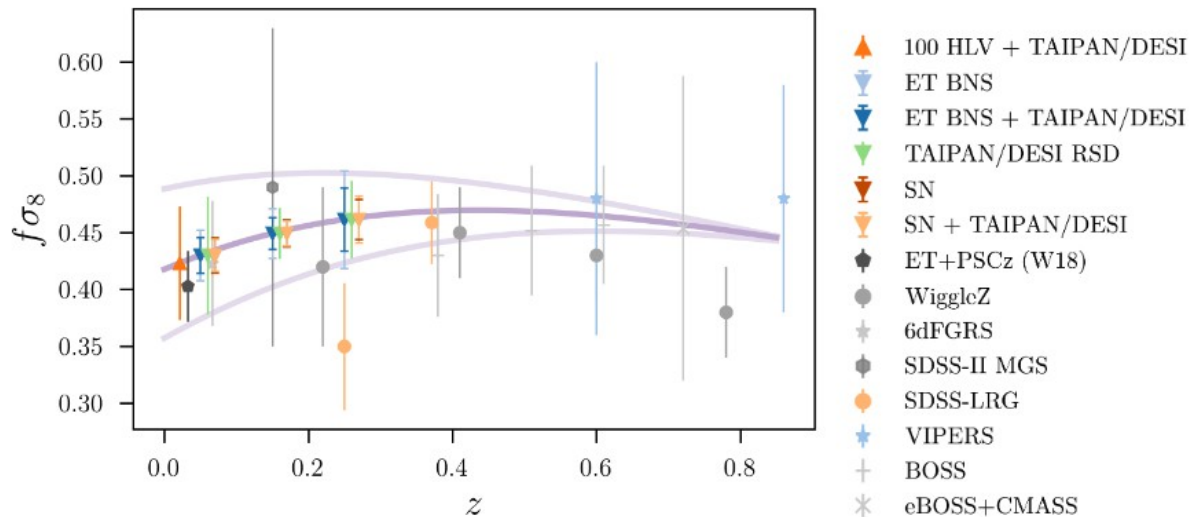
Depends on gravity

$$fD = \Omega_M^\gamma \exp \left(\int_a^1 \Omega_M^\gamma d \ln a \right)$$

Growth rate Linear growth factor

Linder 2005

Growth of structure



- Fix growth index
- 3% precision on $f\sigma_8$ with 3G + galaxy survey (10 yr ET, 5 yr 3 detectors)

Standard Sirens

Schutz 1986

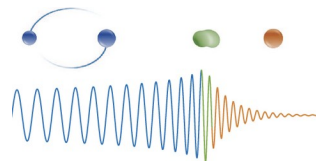
Holz & Hughes 2005

MacLeod & Hogan 2008

Nissanke et al 2010

GW
experiment

d_L



Dimensionless strain

Redshifted chirp mass

frequency

$$h(t) = \frac{M_z^{5/3} f(t)^{2/3}}{d_L} F(\text{angles}) \cos(\Phi(t))$$

Luminosity distance

Position & orientation dependence